QA Test Plan: UDP-Based Zero Touch Provisioning (ZTP)

Field	Value
Document	QA_TEST_PLAN.md (QA persona)
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Companion documents	PRD_ZTP.md (requirement IDs and acceptance criteria) · DESIGN.md (the behaviour under test; constants in Appendix B)
Test basis	PRD v0.1 (requirement IDs), DESIGN v0.1 (§5 wire spec, §8 reliability, §10 auth, §19 demo)

> **Alignment rules.** Every test case (`TC-*`) traces to at least one PRD requirement ID, and every PRD requirement has at least one test case (§10). Constants such as timers, sizes and ports are **not** restated here: tests refer to them by name from DESIGN Appendix B, so a design change can't silently invalidate a test. The demo fixture is the five devices defined in PRD §5.2 and DESIGN §19.3.

Table of contents

1. Scope and objectives
2. Test strategy and levels
3. Test environment and tooling
4. Test data and fixtures
5. Test case catalogue
6. Loss and impairment matrix
7. Security test plan
8. Performance, scale and soak
9. Log and deliverable verification
10. Requirements traceability matrix
11. Entry and exit criteria, defect management, reporting
12. QA timeline, dependencies, assumptions, risks
13. Caveats: what is not or cannot be tested
14. QA enhancement suggestions
15. PDF conformance matrix

1. Scope and objectives

1.1 Objectives

1. Show that every functional, bonus and deliverable requirement in the PRD (§6, §8, §9) is met.

2. Show that the NFR targets are met: reliability under 20% loss, time to OPERATIONAL, scale, robustness and security (PRD §7, §3.3 success metrics SM-1…SM-10).
3. Find defects early in the highest-risk area, the reliability engine (PRD R-01), using deterministic, seeded tests.
4. Produce the evidence the submission needs: logs, the DB dump and a test report.

1.2 In scope

`ztp-device`, `ztp-server`, `libztpcommon`, `mkartifacts`, `run_demo.sh`, and README content.

1.3 Out of scope

Real NAT hardware, real flash or power-loss rigs, IPv6, Windows, and production packaging (PRD §3.2). See §13 for how each gap is mitigated.

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2. Test strategy and levels

Level	What	How	Automated	Owner	When
L1 Unit	Crypto, codec, TLV, reliability engine, util	C test binaries under `tests/`; the reliability engine runs against a **simulated clock and in-memory link**, with no sockets	`make test`	Dev (QA reviews)	Every build, from M1
L2 Component	One binary against a scripted peer	A test harness that sends crafted datagrams (`tests/peer_sim`)	Yes	QA	M2+
L3 Integration	device ↔ server on loopback	Shell scripts `tests/it_*.sh` with netsim flags; assertions on logs, exit codes and the DB dump	Yes	QA	M2+ per milestone
L4 System / E2E	The full demo, scenarios S1–S5	`make demo` plus a results checker	Yes	QA	M6, M7
Negative / security	Bad MAC, replay, tampering, floods, fuzzing	§7	Yes	QA	M3+
Resilience / chaos	Crash points, kill -9, SIGSTOP, rebinding, server restart	Test hooks (DESIGN §19.2) and signals	Yes	QA	M4+
Performance / soak	50+ devices, timing, memory	`make scale`, `/usr/bin/time`, `leaks` or valgrind	Semi	QA	M6, M7
Regression	Everything above	`make test` (every push, CI) and `make release` (the release run, into `results/<…>-release/`)	Yes	QA	M7 and every fix
Documentation review	README coverage	Checklist (§9)	Manual	QA + PM	M7

Principles

- **Determinism first:** every impairment test is run with a fixed `--seed` and recorded with it, so a failure reproduces exactly. Statistical tests run over N seeds (§6).
- **Assert on behaviour, not timing**, wherever possible: counts of retransmissions, states and DB rows, not "within 2.3 s". Timing assertions use generous upper bounds (PRD R-05).

- **Tests run from the logs:** every assertion in L3/L4 uses the tagged log lines (DESIGN §17) or the DB dump. That also checks the logs themselves.

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3. Test environment and tooling

Item	Specification
Platforms	**Local:** macOS 14 on the developer's Mac (Apple clang 16, GNU Make 3.81). **CI:** GitHub Actions `ubuntu-latest` (gcc) and `macos-latest` (clang); see DESIGN §19.4. There is **no local Linux host**, so the Ubuntu runner is the Linux platform for the DEL-02 / NFR-01 sign-off
Network	Loopback (`127.0.0.1`). The broadcast test TC-DISC-05 runs locally on the Mac's Wi-Fi/Ethernet interface (its directed broadcast address) and in CI on the Ubuntu runner
Build variants	`make` (release, `-O2 -Wall -Wextra -Werror`); `make asan` (ASan + UBSan); `make debug` (`-O0 -g`)
Impairment	In-code netsim (`--loss/--reorder/--dup/--delay/--seed`), everywhere; macOS dummynet (commands in DESIGN §16.2; not scripted, TC-SIM-04 waived); Linux `tc netem` (`scripts/netem.sh`), CI Ubuntu runner only, TC-SIM-03
Reduced MTU and real NAT	CI `mtu` job: Linux network namespaces joined by a veth pair at MTU 1280 and 576 (TC-PROV-12/13). Local: `--sim-mtu N:emsgsize` or `N:blackhole` (portable, no sudo). Optional CI `nat-real` job: iptables MASQUERADE with a short conntrack UDP timeout
Capture	`tcpdump -i lo0 -w run.pcap udp` (macOS) or `-i lo` (Linux); Wireshark; `tests/pcapcheck/pcapcheck` (maximum datagram size, fragment filter)
Memory	Locally: ASan + LeakSanitizer (`make asan`) and `leaks --atExit` (built into macOS). CI: valgrind on the Ubuntu runner
Disk footprint (local)	Build outputs, logs, test artifacts and pcaps: < 100 MB in total. `make clean` and `make clean-logs` reclaim it. Large pcaps are limited with `tcpdump -c` . No VMs or containers locally
Fuzzing	The server hook `--self-fuzz N` (`server/fuzz.c` : random and mutated datagrams on a virtual clock, `tests/m6_fuzz.sh`) and the JSON mutation fuzz in the unit tests; libFuzzer was not used
Prerequisite	Xcode Command Line Tools on the macOS host: **verified on 2026-11-10** (Apple clang 16.0.0, GNU Make 3.81; PRD D-08)

QA tools and their options. These options belong to test tooling, not to the ZTP binaries; the binaries' flags are in DESIGN §19.2.

Tool	Options used by the tests	Purpose
`tests/peer_sim`	Modes built in M2: `handshakes N` (TC-SESS-04), `unknown-sid` (TC-SESS-06), `pdf-reorder` (TC-REL-03), `out-of-window` (TC-REL-08), `spoof-source` (TC-NAT-06/12), `rogue-offer` (TC-DISC-07), `replay-offer` (TC-DISC-08), `fake-server` (advertises arbitrary KEEPALIVE_S and DEAD_INTERVALS; TC-CFG-08). Added with their milestones: `rogue-server` , `bad-fleet-key` , `flood N/s` , `replay file.pcap`	A scripted protocol peer for negative, security and component tests (TC-REL-03, TC-AUTH-*, TC-NAT-06/12, TC-CFG-08); driven by `tests/m2_peer.sh`
`tests/pcapcheck/pcapcheck` (built in M2; C, no dependencies)	`--dump-hdr` (decode ZTP headers), `--max-size N` , IP-fragment and seq-gap checks, `--selftest DIR`	Wire-level checks (TC-REL-01, TC-PROV-04/05, TC-APX-03)
`tools/mkartifacts`	`--test-set` (the boundary-size firmware and config fixtures in §4.2); `--fw FILE MODEL VERSION SIZE` (one image of SIZE bytes with a valid header, for tests that upgrade to it)	Deterministic artifacts

Tool	Options used by the tests	Purpose
`tests/check_logs.sh`, `check_appendix_a.sh`, `check_invariants.sh`	`--server` (the server matching rule, DESIGN §17.4)	Log, Appendix A and invariant assertions
valgrind / `leaks`	`--leak-check=full` / `--atExit`	Leak checks (TC-PERF-04)

4. Test data and fixtures

4.1 Device fixture (shared with PRD §5.2 and DESIGN §19.3)

Serial	Model	Start fw	Hooks	Expected final DB row
ZTP-93AF21	EDGE-GW-100	v1.0	—	Success, v1.1, v3, OPERATIONAL
ZTP-84BC19	EDGE-GW-100	v1.0	`--test-hooks --bad-key` (K_mfg corrupted)	Failed, —, —, FAILED, "Invalid HMAC", failed_attempts 3
ZTP-71DD08	EDGE-GW-100	v1.0	`--test-hooks --crash-after-chunks 40 --replay-test`	Success, v1.1, v3, OPERATIONAL; Notes "—" (verbose dump: "Transfer resumed"); attempts 1, failed_attempts 1 (the spliced replay, TC-AUTH-05)
ZTP-55AC90	EDGE-GW-100	v1.1	`--test-hooks --rebind-after 3`	Success, v1.1, v3, OPERATIONAL
ZTP-19FE77	EDGE-GW-050	v0.9	static discovery	Success, v1.0, v2, OPERATIONAL, "Legacy device"

4.2 Artifact fixtures (`mkartifacts --test-set`)

Name	Size	Purpose
`fw_tiny`	1 B	One chunk, minimal
`fw_1023` / `fw_1024` / `fw_1025`	1023 / 1024 / 1025 B	Chunk-boundary checks (1 / 1 / 2 chunks)
`fw_64k`	65,536 B	Standard (64 chunks)
`fw_1m`	1 MiB	Stress (1024 chunks; many CHUNK_REQ batches)
`fw_corrupt`	64 KiB with one flipped byte at serve time (`--corrupt-chunk 17` server hook)	Hash mismatch
`cfg_v3.json` / `cfg_v2.json`	≈ 2.3 KB / ≈ 1.8 KB	Standard configs

Name	Size	Purpose
`cfg_missing_key.json` / `cfg_malformed.json`	—	Validation failures. Committed under `tests/conf/` in M1 together with a fixture `cfg_v3.json` (exactly 2310 B, the DESIGN §11.2 example), so TC-UNIT-17 runs before `mkartifacts` exists

4.3 Policy, key and configuration fixtures (all JSON; DESIGN §25)

- `conf/policy.json` (DESIGN §25.6; assignments only, **no** `nat` **overrides**). `tests/conf/policy_override.json` adds tenant cruise_ops `nat.grace_s = 180` and `serials.ZTP-55AC90.nat.keepalive_interval_s = 10` (TC-CFG-06, TC-SRV-02). `tests/conf/policy_nopolicy.json` adds a key for EDGE-GW-200 with no policy entry (TC-AUTH-12).
- `conf/keys.json` (mode 0600) with the `fleet` key and models EDGE-GW-100 and EDGE-GW-050. The model `EDGE-GW-999` has no key (TC-AUTH-09).
- `conf/server.json`, `conf/device.json` (the production defaults) and the per-side profiles `conf/profiles/server/{demo,strict-nat,cgnat,lan,metered}.json` and `conf/profiles/device/{…}.json` (DESIGN §25.7).
- Negative config fixtures under `tests/conf/`: `bad_range.json` (`nat.keepalive_interval_s: 0`), `bad_type.json` (string where a number is expected), `unknown_key.json` (`nat.keepalive_intervl_s`), `dup_key.json`, `trailing_comma.json`, `too_deep.json` (17 levels), `too_big.json` (> 64 KiB), `dgram_over_cap.json` (`net.max_datagram: 1400`), `chunk_too_big.json` (`xfer.chunk_size: 2048`), `keys_world_readable.json` (mode 0644).
- Two fixtures can't be committed as-is. `too_big.json` (> 64 KiB) is generated by `tests/cli_config.sh` at test time. File modes aren't kept by git, so the key tests copy `tests/conf/keys_test.json` (test-only key material) and set mode 0600 to load it and 0644 for the `keys_world_readable` case.

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5. Test case catalogue

Priority: P1 = must pass for release, P2 = should pass (a waiver needs sign-off), P3 = informational. Each case gives its preconditions and steps in short form, followed by the expected result, which includes the expected **log line** where relevant. All L3/L4 cases run with the release build and the ASan build.

5.1 Unit (L1)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-UNIT-01	SHA-256 known-answer tests	NFR-01, FR-18	P1	FIPS 180-4 vectors: "", "abc", the 448-bit message, 1,000,000 × "a"; streaming across chunk boundaries	All digests match
TC-UNIT-02	HMAC-SHA256 known-answer tests	FR-13	P1	RFC 4231 test cases 1–7, including case 5 (truncation) and long keys	All match

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-UNIT-03	Header and TLV encode/decode round trip	FR-05, BON-02	P1	Every message type × flag combination with random field values; every TLV tag in DESIGN §5.4.1 (including PATH_PROBE 0x1F, RESUMED 0x20, SRV_STAGE 0x21, DEAD_INTERVALS 0x22, MAX_DGRAM 0x23) and every enumeration in §5.4.3	Fields identical after the round trip; `seq` present on every type; header length 44; the resume HELLO_ACK (RESUMED + SRV_STAGE) parses without a duplicate-tag error
TC-UNIT-04	TLV parser bounds	NFR-07	P1	Truncated TLV, length beyond the payload, duplicate tag, unknown tag, zero-length string, oversize string	Unknown tag skipped; duplicate or overrun → reject; never reads out of bounds (ASan)
TC-UNIT-05	Constant-time compare	NFR-06	P2	`ct_memeq` on equal and unequal buffers; check there is no early exit	Correct results; no data-dependent branches
TC-UNIT-06	Serial-number arithmetic and wraparound	FR-05	P1	`seq_lt` across $2^{32} - 1 \rightarrow 0$; the forced re-key threshold	Correct ordering; re-key triggered when `next_seq > 2^30`
TC-UNIT-07	RTO estimator, Karn, backoff	FR-06	P1	Feed RTT samples; mark one frame retransmitted; trigger repeated timeouts	Matches the RFC 6298 formulas; no sample taken from a retransmitted frame; RTO doubles to the cap; clamped to [min, max]
TC-UNIT-08	Reorder buffer: in-order delivery	BON-04	P1	Simulated link: 10,000 seeded random permutations, losses and duplicates of 1..200 frames	The application receives exactly 1..N in order, once each
TC-UNIT-09	Duplicate classification	FR-08, BON-05	P1	Inject seq ≤ rcv_ack, and a seq that's already buffered	Both dropped, counted as dup, and an immediate ACK is scheduled
TC-UNIT-10	Generating and parsing SACK	FR-07	P1	Buffer patterns that produce known bitmaps; sender marks frames sacked	Bit i ⇔ seq ack+2+i; fast retransmit fires after 3 later frames are SACKed
TC-UNIT-11	Policy parser and precedence	FR-29	P1	Model rules, a serial override, default reject, a malformed file	Serial > model > default; a malformed file is a fatal error at startup
TC-UNIT-12	Version comparison	FR-21	P1	v0.9 < v1.0 < v1.1 < v1.10; v1.1 == v1.1; malformed values	Correct ordering; malformed → error

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-UNIT-13	Atomic file write	NFR-10	P1	Inject a failure after `write`, after `fsync`, before `rename`	The file is always the old or the new version, never torn; `.bak` fallback works
TC-UNIT-14	Persisting and loading the chunk bitmap	FR-19	P1	Save a map with arbitrary bits; reload it; mismatched sha	Bits identical; sha mismatch → map discarded
TC-UNIT-15	Netsim determinism and rates	NFR-09, BON-06	P1	Same seed twice; 100k packets at loss 0.15	Identical decision sequences; observed rate 15% ± 1%
TC-UNIT-16	Key-derivation known-answer test	FR-13	P1	Fixed K_fleet, K_mfg, serial, Nd, Ns, sid, epoch	K_boot = HMAC(K_fleet, "ZTP-BOOT-v1"), K_dev = HMAC(K_mfg, "ZTP-DEV-v1" ‖ serial) and K_s equal reference values computed independently (e.g. with `openssl dgst -sha256 -mac HMAC`); inputs encoded as in DESIGN §10.1 (ASCII labels and serial with no terminator, `be64` nonces, `be32` session id, `be16` epoch)
TC-UNIT-17	Config key validation	FR-24	P1	`cfg_v3`, `cfg_missing_key`, `cfg_malformed`	Pass / fail (missing key named) / fail
TC-UNIT-18	Strict JSON parser	FR-34, NFR-07	P1	The JSONTestSuite-style accept/reject corpus (valid, invalid, edge cases), the `tests/conf/` negatives, plus 1M random and mutated inputs (ASan)	Valid files accepted; duplicate keys, trailing commas, comments, depth > 16 and size > 64 KiB rejected with path and line; no crashes or out-of-bounds reads
TC-UNIT-19	Bitmap conversion on chunk-size change	FR-19, FR-20	P1	Maps with random bits at chunk size 1024 converted to 256	Chunk i set at 1024 ⇔ chunks 4i..4i+3 set at 256; the byte ranges are identical; a non-power-of-two ratio is rejected
TC-UNIT-20	Precedence and policy validation	FR-34, FR-35	P1	Synthetic defaults, file, profile, CLI and policy layers; a policy entry with `nat.keepalive_interval_s = 900`	Merge order defaults < file < profile < CLI, then policy tenant < model < serial; the 900 s policy entry is rejected at startup (exit 78)

5.2 Discovery (DISC)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-DISC-01	Soft serial generated on first boot	FR-01	P1	Empty state dir, no `--serial`	`[BOOT] Soft serial generated`, then `[BOOT] Soft serial: ZTP-XXXXXX` matching `^ZTP-[0-9A-F]{6}$`; persisted in `device.state`
TC-DISC-02	Soft serial loaded on later boots, and assigned by configuration	FR-01	P1	(a) Rerun with the same state dir. (b) A fresh state dir with `--serial ZTP-93AF21`. (c) The state dir from (a) with a different `--serial`	(a) `[BOOT] Soft serial loaded`, then `[BOOT] Soft serial: <same value>`; the file is unchanged. (b) `[BOOT] Soft serial assigned (configuration)`. (c) Exit 78 (identity change refused)
TC-DISC-03	Static IP discovery	FR-02	P1	`--discover static --server 127.0.0.1:5555`	`[DISCOVERY] Using static server IP 127.0.0.1:5555`; session established
TC-DISC-04	Multicast discovery	BON-01	P1	`--discover mcast` with no `--server`; `--mcast-iface 127.0.0.1` on both sides (Linux CI: `ip link set lo multicast on`)	`[DISCOVERY] Provisioning server found 127.0.0.1:5555 (OFFER verified)`
TC-DISC-05	Broadcast discovery	BON-01	P2	(a) Locally on the Mac, connected to Wi-Fi/Ethernet: `--discover bcast --bcast-addr $(ifconfig en0 | awk '/broadcast/{print $NF}')`. (b) CI `broadcast` job on the Ubuntu runner	Server found via broadcast on both
TC-DISC-06	Fallback to static	BON-01	P1	Server started with `--test-hooks --no-discovery`; device `--discover mcast --server 127.0.0.1:5555`	`discovery.attempts` (3) DISCOVER attempts, then `[DISCOVERY] Using static server IP 127.0.0.1:5555`; session established
TC-DISC-07	Forged OFFER ignored	BON-01, NFR-06	P1	A rogue responder (`tests/peer_sim rogue-offer`) answers every DISCOVER with a bad-MAC OFFER pointing elsewhere. The server runs with `--test-hooks --no-discovery`, so the rogue is the only responder and the outcome doesn't depend on which reply arrives first	Device logs `[DISCOVERY] OFFER MAC invalid, ignoring` for each, never uses the rogue address, falls back to the static address, and completes with the genuine server
TC-DISC-08	Replayed OFFER ignored	BON-01	P2	`peer_sim replay-offer` captures a genuine OFFER from the server (its own DISCOVER, unicast to the server's discovery port), then replays it to the device's DISCOVERs on a second discovery port	Ignored: `[DISCOVERY] OFFER nonce mismatch, ignoring`; the device falls back to static

5.3 Session (SESS)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-SESS-01	Handshake happy path	FR-04	P1	One device, no impairment	`[UDP] HELLO sent (seq=1)`, `[UDP] Session established (session_id=N)`, N ≠ 0; the server log has the same sid
TC-SESS-02	HELLO lost: idempotent retransmission	FR-06, BON-05	P1	Device `--test-hooks --drop-seq tx:1` (drops the first HELLO transmission)	The retransmission carries the same bytes and Nd; exactly **one** half-open session on the server; the HELLO_ACK cache is hit on any further duplicate
TC-SESS-03	HELLO_ACK lost	FR-06, BON-05	P1	Server `--test-hooks --drop-seq tx:1` (drops its first HELLO_ACK), with `--set rel.rto_initial_ms=2000` so the device's retransmission arrives before the server's own (otherwise either one may recover the handshake, and the cache path isn't exercised)	The device retransmits HELLO; the server answers from its cache (no server-side retransmission of seq 1); one session
TC-SESS-04	Session IDs are random and unique	FR-04	P2	1,000 handshakes	No duplicates among live sessions; values aren't sequential (a simple runs test)
TC-SESS-05	Server down: retry exhaustion and recovery	FR-06, FR-11	P1	Start the device with no server; start the server after 20 s	Retries with backoff and jitter, logged; the device recovers and completes once the server is up
TC-SESS-06	Unknown session_id dropped	NFR-06	P1	`peer_sim` sends (a) a non-HELLO frame with a random sid and (b) a RESUME HELLO with a random sid	(a) Dropped; `rx_unknown` increased; no reply; no state change. (b) Terminal ERROR `SESSION_UNKNOWN` under K_boot; no state change
TC-SESS-07	Stale epoch dropped	FR-23	P1	After a resume (epoch 1), replay an epoch 0 frame	Dropped before the MAC check; `rx_stale` increased
TC-SESS-08	UDP only	FR-03, DEL-04	P1	`lsof -i -a -p <pid>` on both processes during the demo; a pcap filter `not udp` on the test ports; `grep -r SOCK_STREAM src`	Only UDP sockets; no non-UDP packets; no `SOCK_STREAM` in the code
TC-SESS-09	A HELLO can't tear down a live session	FR-14, NFR-06	P1	While ZTP-93AF21 is PROVISIONING, `peer_sim` (holding K_fleet) sends fresh HELLOs for serial ZTP-93AF21, and replays the device's original HELLO	Each gets at most a half-open session or the cached HELLO_ACK; the live session is untouched (same sid, provisioning continues); no `superseded` log line

5.4 Reliability (REL)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-REL-01	Every frame carries a seq	FR-05, BON-02	P1	Decode the pcap of a full run (`build/pcapcheck --dump-hdr`)	Every datagram has a valid header with seq; reliable seqs increase per direction and epoch
TC-REL-02	Retransmission on timeout	FR-06, FR-07, BON-03	P1	Drop one data frame (scripted)	`[UDP] Retransmitting seq=N (timeout)` once; delivered once; the ACK advances
TC-REL-03	The PDF scenario, reproduced verbatim: seq 2 lost, seq 3 buffered, duplicate of seq 2	BON-04, BON-08	P1	**Component test (L2)** against the real server: `peer_sim` (holding the fleet test key) completes a fresh HELLO (device seq 1), then sends path-probe KEEPALIVE frames under K_boot in HALF_OPEN in the order seq 3, seq 2 (with the RETX flag), seq 2 again. This needs only the M2 handshake. (No end-to-end flow reaches this order naturally, because the path probe must be ACKed before REGISTER is sent (DESIGN §8.8.1).)	Server log, in order, **verbatim**: `[UDP] Received packet seq=3 (out of order, expected=2)`, `[UDP] Buffered packet seq=3`, `[UDP] Duplicate packet seq=2 ignored`. The sender-side line `[UDP] Retransmitting seq=2 (timeout)` is checked on the engine in the mirrored unit test (TC-UNIT-08 harness, a simulated clock and a dropped seq 2). Delivery order is 1, 2, 3, each delivered once
TC-REL-04	Lost ACK: data received	BON-07, BON-05	P1	Drop the ACK of REGISTER on the server's TX (scripted: server `--test-hooks --drop-seq tx:2` drops its first REGISTER_ACK, which carries the ACK; with a slower server RTO, `rel.rto_min_ms=rel.rto_initial_ms=1500`, so the device's retransmission arrives before the server's own)	The device retransmits REGISTER (device seq=3, after HELLO=1 and the path-probe KEEPALIVE=2); server logs `Duplicate packet seq=3 ignored`; **one** REGISTER processed (DB attempts = 1)
TC-REL-05	Heavy duplication	FR-08, BON-05	P1	`--dup 1.0` (every datagram duplicated) on both sides	Completes; dup counters ≈ the number of frames; DB rows, stage transitions and chunk writes each happen once
TC-REL-06	Window limit respected	FR-06	P1	Debug log of in-flight counts during a 1 MiB transfer	Never more than `ZTP_TX_WINDOW` unacknowledged
TC-REL-07	Fast retransmit	FR-06	P2	Drop one frame in the middle of the window (scripted)	`Retransmitting seq=N (fast, sack)` before the RTO expires

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-REL-08	Out-of-window frames dropped	NFR-07	P2	`peer_sim` sends seq = rcv_ack + 100 with a valid MAC	Dropped, counted as `out_of_window`, not buffered
TC-REL-09	Completes at 10/15/20% loss	FR-09, NFR-03	P1	Impairment matrix rows M2–M4 (§6), 10 seeds each	100% of the valid devices reach OPERATIONAL
TC-REL-10	Heavy reordering	BON-04	P1	`--reorder 0.25 --reorder-delay 50`	Completes; out-of-order and buffered logs present; no out-of-order delivery to the application (debug assert)
TC-REL-11	No ACK loops	FR-07	P2	Idle session for 60 s with debug logging	ACK and KEEPALIVE_ACK frames are never acknowledged; no ping-pong

5.5 NAT and keepalive (NAT)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-NAT-01	Keepalives when idle	FR-10	P1	An OPERATIONAL device with demo timers, observed for 20 s	`[NAT] Keepalive sent` about every interval; `[NAT] Keepalive acknowledged` for each
TC-NAT-02	No keepalives during a transfer	FR-10	P2	During the `fw_1m` transfer	No KEEPALIVE frames while data is flowing
TC-NAT-03	Device detects a dead server	FR-11	P1	`kill -STOP` the server once the device is OPERATIONAL	After about 3 intervals: `[NAT] Session timeout detected`; resume attempts with backoff; `kill -CONT` → `Session resumed`
TC-NAT-04	Server detects an idle device	FR-11	P1	`kill -STOP` a device	Server: `[NAT] [sid=… …] Session idle, entering grace` after `server_idle_intervals` × keepalive; deleted after `grace_s`; **DB row kept**; a later resume gets `SESSION_EXPIRED` (tombstone)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-NAT-05	NAT rebinding	FR-11, FR-23	P1	`--test-hooks --rebind-after 3`	Server: `[NAT] [sid=… …] Peer address changed A -> B (rebinding)` on the first **new** frame from B; the session continues; no REGISTER; DB attempts unchanged
TC-NAT-06	Spoofed source can't hijack	NFR-06	P1	`peer_sim` sends a frame for a live sid from a new port with a bad MAC	The peer address is unchanged; frame dropped; `mac_fail` increased
TC-NAT-07	No false timeouts under loss (production timers)	FR-11, NFR-03	P2	OPERATIONAL device, **default profile** (keepalive 20 s, dead 3 intervals), 20% loss per direction, 30 min (`make longrun` ; only the OPERATIONAL part counts)	0 session timeouts declared (expected per DESIGN §9.3: P ≈ 6×10^{-7} per cycle, about 90 cycles)
TC-NAT-13	False timeouts with demo timers recover cleanly	FR-11, FR-23	P2	OPERATIONAL device, `demo` profile (2 s / 6 s), 20% loss, 10 min (`make longrun`)	Any `Session timeout detected` is followed by `Session resumed` within `reconnect_max_s` + one RTT; zero re-registrations; the rate is consistent with DESIGN §9.3 (about 6×10^{-3} per cycle)
TC-NAT-08	NAT idle timeout shorter than the keepalive	FR-36, FR-35, FR-11	P1	Demo profile (keepalive 2 s); device `--nat-sim idle_timeout_s=1.5` ; OPERATIONAL for 30 s; then repeat with a policy override `keepalive_interval_s = 1`	Run 1: the session survives (rebinding after each gap), with no re-registration; the server logs `NAT idle timeout is probably below keepalive_interval … consider lowering it` . Run 2, after the override: no more rebindings and no advice lines
TC-NAT-09	NAT reboot: all mappings flushed	FR-36, FR-23	P1	Device `--nat-sim reboot_at_s=T` , once in the middle of the firmware download and once while OPERATIONAL	The server logs `Peer address changed … (rebinding)` ; the transfer continues and completes; the DB has 1 row and attempts = 1
TC-NAT-10	Public IP change	FR-36	P2	Device `--nat-sim change_ip_at_s=T` (Linux CI: 127.0.0.1 → 127.0.0.2; macOS: lo0 alias); run with `nat.allow_ip_change` true, then false	true: server `[NAT] [sid=… …] Peer IP changed A -> B (allowed)` , and the session continues. false: server `[NAT] [sid=… …] Peer IP changed A -> B: RESUME_REQUIRED sent` ; device `[UDP] Resume required by server (IP change)` , then `[UDP] Session resumed (session_id=<same>) epoch=<n>` . Still no re-registration
TC-NAT-11	Outbound UDP blocked by a customer firewall	FR-36, A-09	P1	Device `--nat-sim block_server_port` (and, locally, a real pf block rule on lo0 for port 5555, with sudo); unblock after 60 s	Device log: `[NAT] No response from 127.0.0.1:5555 after 9 HELLO transmissions: check that outbound UDP to port 5555 is allowed (firewall or NAT policy)` ; retries stay at ≤ `net.reconnect_max_s` intervals; after unblocking, provisioning completes normally
TC-NAT-12	A replayed valid frame from a new address can't move the peer	NFR-06, FR-14, FR-11	P1	`peer_sim` captures a valid KEEPALIVE (and a CHUNK_REQ) of a live session and re-sends it from a different port	Classified as a duplicate; **the peer address is unchanged**; the re-ACK goes only to the replayer's address; no `Peer address changed` line; the real device's traffic continues normally

5.6 Authentication and registration (AUTH)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-AUTH-01	REGISTER contents	FR-12	P1	`-v` debug decode of REGISTER, or pcap TLV decode	SERIAL, MODEL, FW_VER, HW_REV, NONCE_REG (64-bit) and NONCE_SRV present
TC-AUTH-02	Valid registration	FR-13	P1	ZTP-93AF21	`[AUTH] Registration successful`, `[AUTH] Assigned tenant: cruise_ops`; DB Success
TC-AUTH-03	Invalid HMAC (wrong model key)	FR-13, FR-30	P1	ZTP-84BC19 `--test-hooks --bad-key` (K_mfg corrupted, K_fleet correct)	HELLO and the path probe succeed; 3 × REGISTER_NAK(BAD_MAC), each echoing NONCE_REG; device logs `[AUTH] Registration failed: Invalid HMAC (1/3)` … `(3/3), giving up` and `[STATE] Device is FAILED (reason: Invalid HMAC)`; DB: Failed, notes "Invalid HMAC", failed_attempts 3; **no** config or firmware sent
TC-AUTH-04	Replay in the same session	FR-14	P1	`--replay-test` phase A	The server logs a duplicate or anti-replay drop; no second DB transition
TC-AUTH-05	Replay spliced into a new session	FR-14	P1	`--test-hooks --replay-test` phase B: the captured REGISTER is re-sent into a second, never-registered half-open session, with its header session_id rewritten (attacker splicing)	The MAC fails, because the header and K_s differ. The server logs an auth failure and NAKs that half-open session. ZTP-71DD08's Success row is **not overwritten**: `failed_attempts` = 1, `attempts` = 1. The device's real session is unaffected
TC-AUTH-06	Reused nonce	FR-14	P1	`--reuse-nonce` (the second registration reuses NONCE_REG with a valid MAC)	REGISTER_NAK(REPLAY)
TC-AUTH-07	Replayed resume HELLO	FR-14, FR-23	P1	`peer_sim` replays the captured resume HELLO (resume_ctr = 2) after the device has resumed with 3	`RESUME_REJECTED`; the live session is unaffected
TC-AUTH-08	Tampering with any byte	NFR-06	P1	For each byte offset of a captured REGISTER and CHUNK_DATA, flip one bit and send	No variant is ever accepted: no REGISTER_ACK, no Success row, no delivered chunk. A tampered REGISTER in a half-open session fails the K_s MAC, which DESIGN §10.3 treats as a verified credential failure (REGISTER_NAK BAD_MAC, a failed attempt recorded, the session discarded); every other tampered frame is dropped silently (`mac_fail`)
TC-AUTH-09	Unknown model	FR-29, FR-13	P1	`--model EDGE-GW-999` (no K_mfg in keys.json; the fleet key is valid)	Terminal ERROR `UNKNOWN_MODEL` (K_boot) in reply to HELLO; no session created; device logs `[AUTH] Registration failed: UNKNOWN_MODEL` and exits FAILED; DB Failed, notes "Unknown model" (capped table, §10.3)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-AUTH-10	HELLO flood is rate-limited	NFR-05	P2	`peer_sim --flood 5000/s` from one IP while 5 devices provision	Rate-limited log lines; all 5 legitimate devices still reach OPERATIONAL
TC-AUTH-11	Rogue server rejected	NFR-06	P1	`peer_sim --rogue-server` answers HELLO with a bad MAC	The device ignores it and keeps retrying with the genuine server
TC-AUTH-12	Known model, no policy	FR-29, FR-13	P1	`keys.json` has a key for EDGE-GW-200, `policy.json` has no entry for it (`default.action = reject`)	REGISTER verified, then REGISTER_NAK `NO_POLICY`; device logs `[AUTH] Registration failed: NO_POLICY` and exits FAILED at once (no retries); DB Failed, notes "No policy"
TC-AUTH-13	Wrong fleet key (not a fleet device)	NFR-06, FR-36	P2	Device with a corrupted K_fleet (`peer_sim --bad-fleet-key`)	Every HELLO silently dropped (`mac_fail` increases on the server); no DB row; the device logs the blocked-path hint after 9 HELLO transmissions and keeps retrying with backoff
TC-AUTH-14	Forged or replayed NAK is ignored	FR-14, NFR-06	P1	During a normal registration, `peer_sim` injects (a) a REGISTER_NAK with a bad MAC and (b) a captured, valid NAK from an earlier attempt (old NONCE_REG)	The device ignores both (MAC check, then nonce check); registration succeeds; the attempt counter is unchanged
TC-AUTH-15	Replayed terminal ERROR is ignored	FR-14, NFR-06	P1	A passive capture of valid K_boot ERRORs (UNKNOWN_MODEL, SERVER_BUSY from one device; SESSION_UNKNOWN and RESUME_REJECTED from a resume) replayed by `peer_sim` to a different device during its HELLO or resume	Each is ignored, because the echoed Nd or Nd′ doesn't match the outstanding request. The target device does **not** enter FAILED, does not re-register, and provisions normally

5.7 Provisioning (PROV)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-PROV-01	Config in chunks with hash	FR-15, FR-17, FR-18	P1	ZTP-93AF21	`Receiving config (chunk 1/3)` … `(3/3)`, `Config checksum verified`

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-PROV-02	Config is valid JSON	FR-15	P1	Parse the saved `config.json` with an external tool (`python3 -m json.tool` or `jq`)	Valid JSON; contains `config_version` and the PDF example's keys (the exact values are checked in TC-PDF-03)
TC-PROV-03	Firmware in chunks with hash	FR-16, FR-17, FR-18	P1	ZTP-93AF21	64 chunks; `Firmware checksum verified` ; the slot file's SHA equals the server artifact's
TC-PROV-04	Datagram size cap	FR-20, NFR-02	P1	pcap of the full demo → maximum UDP payload; also the encoder unit check (`rel_send` returns `-E2BIG` above the current limit)	≤ 1200 bytes (expected maximum 1076); after a forced fallback (TC-PROV-13) ≤ 508
TC-PROV-05	DF set, no IP fragments	FR-20	P1	Linux and macOS: check the socket option (`getsockopt IP_MTU_DISCOVER` = `IP_PMTUDISC_DO` / `IP_DONTFRAG` = 1); pcap filter `ip[6:2] & 0x3fff != 0` on the reduced-MTU CI link (TC-PROV-12/13)	DF set on both platforms; 0 fragments, including on a 1280 and a 576 MTU link (on loopback, MTU 16 KB, a zero-fragment result alone proves nothing, so the reduced-MTU links are the evidence)
TC-PROV-06	Chunk boundaries	FR-17	P1	`fw_tiny` , `fw_1023` , `fw_1024` , `fw_1025` , `fw_1m`	Chunk counts 1, 1, 1, 2, 1024; all hashes verified. The 1-byte image is too small for a firmware header (DESIGN §11.1), so after its hash verifies the header check rejects it: `Firmware header does not name …` , then FAILED (HASH_MISMATCH, §12.6); the four others reach OPERATIONAL. The test images carry valid headers (`mkartifacts --fw`)
TC-PROV-07	Hash mismatch handling	FR-18	P1	The artifact differs from its manifest: one byte of the firmware file is changed on disk after the server has hashed it at startup, so every download disagrees with the manifest (the planned `--corrupt-chunk 17` hook is equivalent and not needed)	`Checksum mismatch, discarding (attempt 1/2)` , again for 2/2, then STATUS_REPORT FAILED HASH_MISMATCH; DB stage FAILED
TC-PROV-08	Resume after kill -9	FR-19	P1	ZTP-71DD08 `--crash-after-chunks 40` , plus a separate run with an external `kill -9` at a random time	After the restart: `Resuming firmware: 32/64 chunks present, requesting 32` ; the total number of chunks served is < 64 + 32; the hash verifies

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-PROV-09	Resume after a server restart	FR-19, FR-23	P1	Restart the server (SIGTERM, then start) in the middle of the firmware download	The server restores the session from the journal into IDLE_GRACE; the device **resumes** (no second `[AUTH] REGISTER:` line; server `Registration successful` count stays 1); the partial download is reused (manifest sha matches)
TC-PROV-10	Artifact changed between attempts	FR-18, FR-19	P2	Replace the firmware artifact (new sha) while the device is down	The partial is discarded; full download; verified
TC-PROV-11	Duplicate chunks are harmless	BON-05	P1	`--dup 0.5` on the server's TX during the firmware download	No rewrites (write counter = chunk count); the duplicates are caught by the reliability layer as duplicate seqs (device `dup` counter > 0), so none reaches the application; hash verifies
TC-PROV-12	1280-byte MTU path: no fallback needed	FR-20, NFR-02	P1	CI `platform-linux`: `lo` MTU set to 1280, one full provisioning (`tests/check_appendix_a.sh` with `APX_SEEDS=0 APX_NO_STEPDOWN=1`), captured and checked by `pcapcheck --max-size 1200` (a veth pair adds nothing here: the limit is the interface MTU either way)	Provisioning completes at 1200 B datagrams; no step-down log; 0 fragments
TC-PROV-13	Small-MTU path: step-down to 508 B	FR-20, FR-19	P1	(a) CI `mtu` job at veth MTU 576 (kernel EMSGSIZE on the server). (b) Locally, **server** `--sim-mtu 1000:emsgsize`. (c) Locally, **server** `--sim-mtu 1000:blackhole` (large server frames silently dropped). Each is applied in the middle of the firmware download	(a, b) Server: `[UDP] [sid=… …] Path MTU too small for 1200 B (EMSGSIZE): datagram limit now 508`; the stuck chunk seqs are resent as 52 B fill frames with `MTU_REDUCED`. (c) Device: `[UDP] Suspected PMTU black hole: datagram limit now 508` after `net.pmtu_blackhole_s`, then MANIFEST_REQ with MAX_DGRAM 508; the server fills its stuck seqs. In every case: CHUNK_SIZE 256 in the new MANIFEST, the chunks already received are **kept** (bitmap converted), no seq gap or stall, the hash verifies, and the max datagram is ≤ 508 from then on, including after a resume

5.8 Upgrade and restart (UPG)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-UPG-01	Outdated firmware upgraded	FR-21	P1	ZTP-93AF21 (v1.0)	`[UPGRADE] Current: v1.0 -> Target: v1.1`; `Upgrade complete, restarting device`
TC-UPG-02	Process restart	FR-22	P1	Both `--restart-mode exec` and `--restart-mode exit` (supervised); binary started via a `PATH` lookup (not an absolute path)	`[BOOT] Device starting (boot #2)` then `[BOOT] Restart detected (pending upgrade to v1.1)`; boot_count increased; the exec path is resolved correctly (macOS `_NSGetExecutablePath`); no leaked fds (`lsof` before and after shows the same count)
TC-UPG-03	Resume without re-registration	FR-23	P1	After TC-UPG-01	`[UDP] Session resumed (session_id=<same>) epoch=1`; the device log has exactly **1** `[AUTH] REGISTER:` line; the server log has exactly 1 `Registration successful` for the serial (server matching rule, DESIGN §17.4); the server logs a peer address change (new socket, so a new NAT mapping)
TC-UPG-04	One DB row per serial	FR-23, FR-30	P1	After the full demo	The dump has exactly one row per serial; attempts = 1 for every device except the bad-key one
TC-UPG-05	Already current: no upgrade	FR-21	P1	ZTP-55AC90 (v1.1)	`Firmware v1.1 is current, no upgrade required`; no restart; no firmware chunks transferred
TC-UPG-06	Legacy device path	FR-29, FR-21	P1	ZTP-19FE77 (EDGE-GW-050, v0.9)	Upgrades to v1.0; cfg v2; note "Legacy device"
TC-UPG-07	Crash at each upgrade step	NFR-10	P1	`--test-hooks --crash-at {download,downloaded,verified,committed,post-exec}`: all 5 points in DESIGN §12.4, run one at a time	Each case recovers to OPERATIONAL with the correct firmware; no re-download after `verified`; no duplicate registration
TC-UPG-08	Grace period expired before the resume	FR-23	P1	`--test-hooks --restart-mode exit --restart-delay <grace_s + 5>`	Resume → terminal ERROR `SESSION_EXPIRED`; fresh registration; **same** DB row; provisioning note (verbose dump) "Re-registered after session expiry"; the default dump's Notes are unchanged

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-UPG-09	Rollback on failed validation	FR-25	P2	`--fail-validate`	`Validation failed on v1.1, rolled back to v1.0`; STATUS FAILED VALIDATION_FAILED; DB registration Success, stage FAILED
TC-UPG-10	Fast restart while the server session is still live	FR-23, FR-19	P1	ZTP-71DD08 `--test-hooks --crash-after-chunks 40` relaunched within 1 s (before any idle timeout)	The server is still in PROVISIONING when the resume HELLO arrives; resume accepted (epoch + 1), old TX state discarded; the device re-requests the missing chunks; DB attempts = 1
TC-UPG-11	Crash during resume, then resume again (no deadlock)	FR-23, NFR-10	P1	(a) Device `--test-hooks --crash-at resume-sent` (abort right after the resume HELLO is sent; the server may or may not have accepted it). (b) Server SIGKILL right after it sends HELLO_ACK(RESUMED), restarted at once	(a) After the relaunch, the device proposes `proposed_epoch = max(epoch, proposed_epoch) + 1` with a higher ctr; the server accepts **on the first try**, whether or not it had accepted the earlier proposal; the device logs `Session resumed (session_id=<same>) epoch=<n>`. (b) The device retransmits its resume HELLO (the same ctr and Nd′); the restarted server, which journaled the accept with Nd′ and Ns2 before replying, rebuilds and resends the identical HELLO_ACK (rule 1). No re-registration in either case; DB attempts = 1

5.9 Validation (VAL)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-VAL-01	Happy-path validation	FR-24..27	P1	ZTP-93AF21	`Configuration present`, `Firmware v1.1 verified`, `Session healthy`, `[STATE] Device is OPERATIONAL`
TC-VAL-02	Config missing at validation	FR-24	P1	`--test-hooks --hook-delete-config` (then again with `--no-refetch`)	One re-fetch of the config, then pass; with `--no-refetch` → STATUS_REPORT FAILED and `[STATE] Device is FAILED (reason: …)`
TC-VAL-03	Firmware version mismatch	FR-25	P1	(a) After an upgrade: `--test-hooks --hook-bad-fw-version v9.9`. (b) A device whose firmware is already current, same hook	(a) Rollback to slot A, `[UPGRADE] Validation failed on v1.1, rolled back to v1.0`, FAILED. (b) FAILED with no rollback (nothing to roll back)
TC-VAL-04	Unhealthy session	FR-26	P1	`kill -STOP` the server at the start of VALIDATE (`--test-hooks --hook-hold-health-probe 2` holds the probe so the stop lands there); (a) `val.deadline_s=25`, SIGCONT after 8 s; (b) `val.deadline_s=8`, never resumed	Not OPERATIONAL until the session recovers; `kill -CONT` within the deadline → OPERATIONAL; past it → FAILED
TC-VAL-05	Server records OPERATIONAL	FR-27, FR-30	P1	After TC-VAL-01	Server `[STATE] … OPERATIONAL`; DB stage OPERATIONAL

5.10 Server (SRV)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-SRV-01	5-device concurrent demo	FR-28	P1	`make demo`	Every final DB row matches §4.1
TC-SRV-02	Per-device assignment	FR-29	P1	`tests/m6_server.sh` : a policy whose serial entry for ZTP-55AC90 assigns cfg v2 and a 10 s keepalive under a tenant grace of 180 s, beside the model entries (the `tests/conf/policy_override.json` fixture overrides NAT values only)	Each device gets the firmware and config of its model, or of its serial override
TC-SRV-03	DB record fields	FR-30	P1	Inspect the dump and `db.jsonl`	Device ID, timestamp, firmware assigned, config version and outcome present for every row
TC-SRV-04	Dump on demand	FR-31, DEL-11	P1	`kill -USR1 <server>` during the demo; SIGINT at the end	The dump is printed both times, in the default PDF format (exact format checks in TC-PDF-02); `--dump-verbose` adds the extra columns **after** Notes only
TC-SRV-05	Summary	FR-32	P1	At the end of the demo	Lists 4 successes, 1 failure with reason, and firmware/config per device
TC-SRV-06	DB persists across a server restart	NFR-10	P1	Restart the server after the demo; dump	The same rows are rebuilt from the journal
TC-SRV-07	Fuzzing malformed datagrams	NFR-07	P1	1M random and mutated datagrams into the control and discovery handlers, ASan build (`tests/m6_fuzz.sh` : the server hook `--self-fuzz` , in process on a virtual clock so the rate limiter does not discard most inputs and timers fire; 3 seeds × 334,000 in the CI `asan` job)	No crash, hang or sanitizer report; counters increase
TC-SRV-08	Session limits and half-open expiry	NFR-05, NFR-07	P2	300 half-open HELLOs from `peer_sim` spread over 30 source addresses (127.0.0.10–39 on Linux; or `--set limits.hello_rate_per_ip=1000`) so the rate limit doesn't mask the cap	Capped at `max_half_open` (256); further HELLOs get terminal `SERVER_BUSY` ; they expire after `half_open_ttl_s` ; memory returns to baseline
TC-SRV-09	Interleaved progress	FR-28	P1	Server log of the demo	Lines from ≥ 3 different sids interleave during provisioning (not one device after another)
TC-SRV-10	Resume across a server restart mid-upgrade	FR-23, NFR-10	P1	ZTP-93AF21 with `--test-hooks --restart-mode exit --restart-delay 5` (a window while the server is in AWAIT_RESUME); during that window, SIGKILL the server and restart it	Server log: session 7421 restored into IDLE_GRACE from the journal; the device resumes with `[UDP] Session resumed (session_id=7421) epoch=1` ; no second REGISTER; DB row unchanged

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-SRV-11	Failed-record cap	NFR-07, NFR-05	P2	`peer_sim` (holding K_fleet) performs 20 000 failing registrations with random serials; `--set limits.max_failed_records=10000`	The DB never holds more than 10 000 Failed rows without a prior Success (oldest evicted); Success rows are never evicted or overwritten

5.11 Simulation (SIM)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-SIM-01	Observed loss rate, per direction and end to end	BON-06	P1	10k datagrams each way, `--loss 0.15` on both processes	Each sender's `[SIM]` drop rate is 15% ± 2%, and the end-to-end loss per direction computed from both processes' counters (sent by A, received by B) is 15% ± 2%, not about 28% (TX-only rule, DESIGN §16.1)
TC-SIM-02	Reordering and duplication are observable	BON-06	P1	`--reorder 0.1 --dup 0.05` (the M2 run uses 0.5 / 0.5, because a handshake has few frames)	`[SIM]` counters non-zero; matching dup events in `[UDP]`. OOO events need two reliable frames in flight in one direction, which a handshake never has (HELLO and the path probe are sequential), so the OOO half is checked from M4 on, during transfers
TC-SIM-03	The netem alternative	BON-06	P2	CI `impair-netem` job (Ubuntu runner): netem on `lo` (inline in the workflow), netsim off, run the demo	Same PASS result as with netsim. *If `sch_netem` can't be loaded on the hosted runner, this is waived (PRD R-11)*
TC-SIM-04	The macOS dummynet alternative	BON-06	P2	**Waived at M7** (needs `sudo` to replace the pf rules; not run; TC-SIM-03 and the in-code netsim cover BON-06). Planned: locally, `sudo scripts/dummynet.sh on 0.15 5` (loss and delay only, **both directions**), netsim off, run the demo; then `scripts/dummynet.sh off` and check that the pf rules were restored	Same PASS result; retransmission counters show OS-level loss in **both** directions (device and server `retx_timeout` > 0); reordering and duplication aren't exercised (a dummynet limitation)

5.12 Performance (PERF)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-PERF-01	50 concurrent devices	NFR-05, FR-28	P1	`make scale N=50` (`scripts/run_scale.sh`: 4 in 5 devices EDGE-GW-100 upgrading, 1 in 5 legacy), 15% loss; also in the CI `impair-matrix` job	All valid devices OPERATIONAL; all done ≤ 120 s; server peak RSS ≤ 64 MB
TC-PERF-02	Time to OPERATIONAL	NFR-04	P1	20 runs × the 5-device demo at 20% loss	Median ≤ 60 s, p95 ≤ 90 s
TC-PERF-03	Retransmission overhead against the model	FR-09	P3	Counters at 10/20/30% loss	Ratio of transmissions to frames ≤ DESIGN §21.1 Q5 bounds (+10%)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-PERF-04	Soak with no leaks	NFR-07	P2	5 OPERATIONAL devices for 1 h (`make soak` , `scripts/run_soak.sh` ; `leaks` on the live processes on macOS; the CI `valgrind` job runs 3 min under memcheck)	No leaks; RSS stable (± 5%). On macOS the check is growth of at most 5%: the kernel compresses an idle process's pages, so resident size *falls* over a soak (seen: 30–60%), and a fall is not a leak

5.13 Logs (LOG)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-LOG-01	Required tags present	FR-33	P1	grep the demo logs	Each of `[BOOT] [DISCOVERY] [UDP] [NAT] [AUTH] [PROVISION] [UPGRADE] [VALIDATION] [STATE]` appears in at least one device log, and the relevant ones in the server log
TC-LOG-02	Bonus log lines present in the demo	BON-08	P1	grep the demo logs with the **generic** pattern (real sequence numbers vary by seed): `Received packet seq=[0-9]+ \(out of order, expected=[0-9]+\)` , `Buffered packet seq=[0-9]+` , `Retransmitting seq=[0-9]+ \(timeout\)` , `Duplicate packet seq=[0-9]+ ignored`	All four lines appear; the literal PDF numbers (seq 2 and 3) are proven by TC-REL-03
TC-LOG-03	Format and files	NFR-08	P2	Inspect	ISO-8601 timestamps with ms, monotonic within each file; one file per device; the device log continues across the restart
TC-LOG-04	Counters at exit	NFR-08	P2	End of the demo	Counter summary present on both sides

5.14 Deliverables (DEL)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-DEL-01	Clean build on both platforms	DEL-02, NFR-01, NFR-11	P1	Fresh clone → `make` on macOS and Linux	Builds with no warnings (`–Werror`); links only libc
TC-DEL-02	The demo produces its artifacts	DEL-01, DEL-09, DEL-10, DEL-11	P1	`make demo`	Exit 0, PASS; `logs/server.log` , five `logs/device_*.log` , `logs/db_dump.txt`
TC-DEL-03	Device/server separation	DEL-03	P1	Directory layout check; `nm` symbol check; include check	No server symbols in the device binary and vice versa; no cross-includes
TC-DEL-04	The README quick start works	DEL-05	P1	A reviewer who has never seen the project follows the README only	Builds and runs the demo successfully

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-DEL-05	README: 5 required answers	DEL-06	P1	§9 checklist	All 5 present and accurate against the implementation
TC-DEL-06	README: design sections	DEL-07	P1	§9 checklist	All 6 topics present
TC-DEL-07	README: bonus coverage	DEL-08	P1	§9 checklist	3 bonus topics + 4 bonus answers present
TC-DEL-08	Sanitizer-clean demo	NFR-11	P1	`make asan demo`	No ASan or UBSan reports
TC-DEL-09	Logs committed	DEL-09, DEL-10	P1	`git ls-files logs/` (in the submission)	The server and every client log are present
TC-DEL-10	README: why keepalives matter	DEL-12	P1	§9 checklist, item 18	A dedicated README section explains NAT/firewall idle expiry, why the server can't initiate, failure detection, resource hygiene and rebinding (DESIGN §9.6), and agrees with the implemented timers

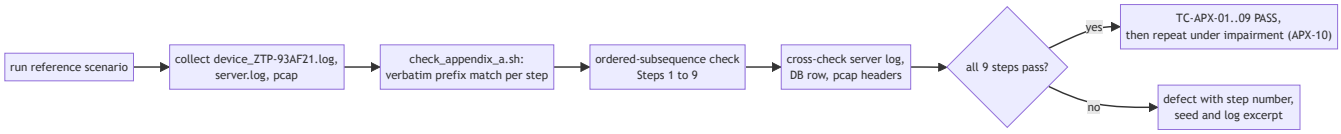
5.15 End-to-end (E2E)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-E2E-01	Appendix A flow, single device	FR-33, FR-01..27	P1	ZTP-93AF21 alone, no impairment, 127.0.0.1 (runs everywhere, including CI)	All of TC-APX-01..09 pass, and the PDF lines appear in Appendix A order (an ordered-subsequence check; §5.17)
TC-E2E-02	Full demo under impairment	FR-28..32, NFR-03	P1	`make demo` (15% loss, 10% reorder, 5% dup) × 10 seeds	PASS for all seeds; the dump matches §4.1

5.17 Conformance with PDF Appendix A (APX), one test per step

Setup (the reference run), the same as DESIGN Appendix A: `sudo ifconfig lo0 alias 10.10.10.5/32`; server `--bind 10.10.10.5 --profile demo --test-hooks --first-session-id 7421`; device `--serial ZTP-93AF21 --model EDGE-GW-100 --fw v1.0 --discover static --server 10.10.10.5:5555 --profile demo --test-hooks --nonce 984312`; no impairment (TC-APX-01..09) and 15% loss, 10% reorder, 5% dup (TC-APX-10). On CI (no sudo needed for an alias, but a fixed IP isn't guaranteed) the same tests run against `127.0.0.1`, and only the address is substituted.

Matching rule, from DESIGN §17.4: each PDF line must appear **verbatim as the prefix** of a log message (`grep -F`). Order is checked as an ordered subsequence across the whole device log. Server-side evidence is checked in `server.log` for the same sid. **Server lines** carry the session tag after the log tag (`[AUTH] [sid=7421 ZTP-93AF21] Registration successful`), so `check_appendix_a.sh` matches them with the server rule from DESIGN §17.4: remove the `[sid=… …]` part, then prefix-match.



TC ID	PDF Appendix A step	Req	Pri	Check	Expected result: the PDF's text, verbatim
TC-APX-01	Step 1: Device boot (device information: ZTP-93AF21, EDGE-GW-100, v1.0)	FR-01, FR-33	P1	Device log	`[BOOT] Device starting` → `[BOOT] Soft serial: ZTP-93AF21` → `[BOOT] Firmware version: v1.0`
TC-APX-02	Step 2: Server discovery (static IP 10.10.10.5, UDP port 5555)	FR-02, FR-33	P1	Device log; pcap (`tcpdump -i lo0 udp and host 10.10.10.5`)	`[DISCOVERY] Using static server IP 10.10.10.5:5555` ; the pcap shows the device's datagrams addressed to 10.10.10.5:5555
TC-APX-03	Step 3: UDP session (HELLO seq=1 with soft_serial; HELLO_ACK seq=1 with session_id=7421)	FR-03, FR-04, FR-05, FR-33	P1	Device log + pcap header decode (`build/pcapcheck --dump-hdr`)	`[UDP] HELLO sent (seq=1)` → `[UDP] Session established (session_id=7421)` . The pcap shows HELLO with seq=1 and the SERIAL TLV `ZTP-93AF21` , and HELLO_ACK with seq=1 and session_id=7421
TC-APX-04	Step 4: NAT keepalive	FR-10, FR-33	P1	Device log order	`[NAT] Keepalive sent` → `[NAT] Keepalive acknowledged` , appearing **after** Step 3 and **before** Step 5 (the path probe); periodic keepalives continue after OPERATIONAL
TC-APX-05	Step 5: Secure registration (session_id=7421, soft_serial, model, firmware v1.0, nonce=984312)	FR-12, FR-13, FR-29, FR-33	P1	Device log, server log, DB	`[AUTH] REGISTER: session_id=7421 soft_serial=ZTP-93AF21 model=EDGE-GW-100 firmware=v1.0 nonce=984312` , then `[AUTH] Registration successful` → `[AUTH] Assigned tenant: cruise_ops` . The server logs the same; the DB row is Success
TC-APX-06	Step 6: Config provisioning (the example config, 3 chunks)	FR-15, FR-17, FR-18, FR-33	P1	Device log; saved config (with TC-PDF-03)	`[PROVISION] Receiving config (chunk 1/3)` → `(chunk 2/3)` → `(chunk 3/3)` → `[PROVISION] Config checksum verified`
TC-APX-07	Step 7: Firmware upgrade (v1.0 → v1.1)	FR-16, FR-21, FR-22, FR-33	P1	Device log; slot B header	`[UPGRADE] Current: v1.0 -> Target: v1.1` → `[UPGRADE] Upgrade complete, restarting device` ; slot B header = v1.1
TC-APX-08	Step 8: Resume after restart, without duplicate registration	FR-22, FR-23, FR-33	P1	Device log; server log; DB	`[BOOT] Restart detected` → `[UDP] Session resumed (session_id=7421)` . **No** `[AUTH] REGISTER:` line after the restart; the server has exactly 1 `Registration successful` for ZTP-93AF21; the DB has one row with attempts = 1

TC ID	PDF Appendix A step	Req	Pri	Check	Expected result: the PDF's text, verbatim
TC-APX-09	Step 9: Validation and operational mode	FR-24..27, FR-33	P1	Device log; server log; DB	`[VALIDATION] Configuration present` → `[VALIDATION] Firmware v1.1 verified` → `[STATE] Device is OPERATIONAL`; the server's DB stage is OPERATIONAL
TC-APX-10	Steps 1–9 under impairment	FR-09, NFR-03, FR-33	P1	Same checks, with 15% loss, 10% reorder and 5% dup × 10 seeds; **a fresh server and state directory per seed** (so the fixed `--nonce 984312` never hits the nonce cache)	All step checks of APX-01..09 still pass in order; extra retransmission, out-of-order or duplicate lines may appear in between

5.18 Conformance with the PDF's other examples and its scope (PDF)

TC ID	PDF item	Req	Pri	Steps / setup	Expected result
TC-PDF-01	§1 "Example output"	FR-01, FR-02, FR-33	P1	First boot (empty state dir, no `--serial`), once in static and once in multicast mode	Both runs contain `[BOOT] Soft serial generated` and `[DISCOVERY] Provisioning server found`, in that order
TC-PDF-02	§9 "Example of a Dump from Server DB"	FR-31, FR-30, FR-32, DEL-11	P1	Full demo, then `logs/db_dump.txt`	Title `Device Registration Details`; header columns exactly `Device ID, Registration Status, Firmware Assigned, Config Version, Registration Time, Notes` in that order; times match `^\d{4}-\d{2}-\d{2} \d{2}:\d{2}:\d{2}$`; `—` in empty cells. The rows match the PDF's 5 example rows: ZTP-93AF21 Success v1.1 v3 —; ZTP-84BC19 Failed — — Invalid HMAC; ZTP-71DD08 Success v1.1 v3; ZTP-55AC90 Success v1.1 v3; ZTP-19FE77 Success v1.0 v2 Legacy device: **an exact match apart from the timestamps**. Rows are sorted by Registration Time, then Device ID (DESIGN §14.5), so the test compares the set of rows **and** checks that sort order; rows exist only for verdicts (DESIGN §14.4), so no half-finished registrations appear. Provisioning notes such as "Transfer resumed" appear only in `--dump-verbose`
TC-PDF-03	Appendix A "Example Configuration"	FR-15	P1	Parse the received `config.json` with `jq` (or `python3 -m json.tool`)	`.interfaces.wan == "eth0"`, `.interfaces.lan == "eth1"`, `.mtu == 1500`, `.keepalive_interval == 20`; extra keys allowed
TC-PDF-04	Appendix A addressing (10.10.10.5:5555)	FR-02	P3	Local Mac with the lo0 alias (requires sudo); the alias is removed afterwards	The reference run works on the exact PDF address; logs show `10.10.10.5:5555`

TC ID	PDF item	Req	Pri	Steps / setup	Expected result
TC-PDF-05	Overview / Scope & Expectations: UDP only, "not a REST or API exercise", C or Python, no UI	FR-03, NFR-01, DEL-04	P1	`git ls-files` (only `.c`, `.h`, `.sh`, Makefile, docs); `grep -rniE 'http|SOCK_STREAM|listen\(|accept\('` over the code; `nm` of both binaries	C sources only; no HTTP, REST or TCP code; no listening TCP sockets at runtime (`lsof -iTCP` empty for both processes); interaction is CLI and logs only

5.19 Configuration (CFG)

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-CFG-01	Built-in defaults	FR-34	P1	Start both binaries with no config file, then run `--print-config`	Values equal DESIGN Appendix B.2 defaults; every value's source is `default`
TC-CFG-02	A config file overrides the defaults	FR-34	P1	`server.json` setting `nat.keepalive_interval_s = 15`, `rel.tx_window = 4`; `device.json` setting `rel.tx_window = 4`	`--print-config` on both shows the values with source = the file; in a session, the server advertises KEEPALIVE_S = 15 and the device sends keepalives every 15 s; at most 4 frames are in flight (debug log)
TC-CFG-03	Precedence: defaults < file < profile < CLI	FR-34	P1	`server.json` setting keepalive 5, plus `--profile demo`, plus `--set nat.keepalive_interval_s=3`; check `--print-config` after each removal	Effective keepalive = 3 (source `cli`); removing `--set` gives 2 (`profile:demo`, which overlays the file); removing `--profile` gives 5 (file); removing the file gives 20 (`default`)
TC-CFG-04	Range and type validation (fail fast)	FR-34, NFR-07	P1	Each `tests/conf/bad_*.json`, `dgram_over_cap.json`, `chunk_too_big.json`	Exit code 78 with a message naming the key, value, file and allowed range; nothing is bound on the network
TC-CFG-05	Unknown keys rejected (typo protection)	FR-34	P1	`unknown_key.json`, then the same with `--config-lenient`	Strict: exit 78, `unknown key nat.keepalive_intervl_s`. Lenient: a warning, and the default is used

TC ID	Title	Req	Pri	Steps / setup	Expected result
TC-CFG-06	Per-tenant, model and serial NAT override delivered to the device	FR-35, FR-29	P1	Server with `--policy tests/conf/policy_override.json` (tenant cruise_ops `grace_s` 180; ZTP-55AC90 `keepalive_interval_s` 10); default profile; the 5-device demo	Server log `Policy: … keepalive 10 s … (source: policy:serial ZTP-55AC90)`; HELLO_ACK to ZTP-55AC90 carries KEEPALIVE_S = 10 and to the others 20; every HELLO_ACK carries GRACE_S = 180 (tenant); ZTP-55AC90 sends keepalives every 10 s, the others every 20 s; the verbose dump shows the effective values
TC-CFG-07	Profiles	FR-34	P2	Each of `demo`, `strict-nat`, `cgnat`, `lan`, `metered`: `--profile <name> --print-config` on the server (loads `conf/profiles/server/<name>.json`) and on the device (`conf/profiles/device/<name>.json`)	Each side's file passes strict validation (exit 0, no unknown keys); values match DESIGN §25.7; `--demo-timers` is the same as `--profile demo`
TC-CFG-08	Device clamp on advertised values	FR-35, NFR-06	P1	`peer_sim fake-server` (the real server would reject these values, §25.8) advertises KEEPALIVE_S = 1 to a device with `keepalive_min_s = 5`, then KEEPALIVE_S = 900 against `keepalive_max_s = 300`, then DEAD_INTERVALS = 50. The device bounds are set in a config file, not with `--set`: a device-side `--set nat.keepalive_interval_s` overrides any advertised value by design (DESIGN §25.3)	The device uses 5 s, then 300 s, then 10 intervals, and logs `[NAT] Advertised keepalive 1 s clamped to 5 s` (and so on); no flooding and no silence
TC-CFG-09	Key-file permissions, and policy/artifact checks	FR-34, NFR-06	P1	`keys.json` mode 0644; a policy referencing a missing `firmware_file`; a model without a key	Each refuses to start (exit 78) with a specific message
TC-CFG-10	Settings from the resolved decisions take effect (DESIGN §23)	FR-34, FR-35	P2	One run per setting: (a) `run.exit_when_operational` false, then true; (b) `rel.rto_initial_ms=1000 rel.rto_min_ms=1000 rel.rto_max_ms=60000` (the RFC 6298 values); (c) `discovery.mode` static, then bcast; (d) `log.dump_verbose` true; (e) `auth.register_attempts=1` with `--bad-key`; (f) `nat.allow_ip_change=false`; (g) `net.mtu_fallback=false` with `--sim-mtu 1000:emsgsize`	(a) false: the device stays OPERATIONAL and keeps sending keepalives; true: it logs `[STATE] Exiting after operational hold (run.exit_when_operational, <h> s)` and exits 0 after `run.operational_hold_s`. (b) Accepted by validation; `--print-config` shows it; the first retransmission waits ≥ 1 s. (c) The chosen discovery path is logged. (d) `db_dump_verbose.txt` written, default dump unchanged. (e) FAILED after one NAK. (f) Behaves as TC-NAT-10 false. (g) No step-down: the transfer fails visibly with `-E2BIG`/EMSGSIZE logged, so the switch really disables the fallback. Each is also visible in `--print-config`

Total: 165 catalogued test cases (P1: 139, P2: 24, P3: 2), plus the exploratory TC-PERF-01b (§8).

5.20 Where each test runs

Environment	What runs there
Local Mac (developer, daily)	All of L1, all of L3/L4 with netsim, the TC-APX reference run on 10.10.10.5 (lo0 alias) and TC-PDF-04, TC-DISC-05 (a) (Wi-Fi broadcast), TC-SIM-04 (dummynet), the pcap cases, the performance cases (TC-PERF-01/02: timing is measured locally, not on shared runners), `leaks` /ASan soak
CI `macos-latest`	Clean build, `make test`, demo, ASan, impairment rows M1–M4 (a second, clean macOS environment), TC-APX-01..10 and TC-PDF-01/02/03/05 on 127.0.0.1
CI `ubuntu-latest` (the only Linux)	gcc build, `make test`, demo, ASan, rows M1–M4, TC-SIM-03 (netem), TC-DISC-05 (b) (broadcast), valgrind (TC-PERF-04, Linux half), row M7, TC-APX-01..10 and TC-PDF-01/02/03/05 on 127.0.0.1

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6. Loss and impairment matrix

Loss, reorder and duplication are configured **per direction**: they are applied at each sender only (DESIGN §16.1), so "15%" means 15% of the datagrams in each direction are lost, not a compounded 28%. The 5-device demo is run against each row, with **10 seeds per row** (seeds 1–10), on the local Mac and both CI runners for rows M1–M4, on the local Mac for M5–M6 and M8–M9, and on the CI Ubuntu runner only for M7 (netem). CI runs 3 seeds per push and all 10 nightly or on demand.

Row	Loss (each direction)	Reorder	Dup	Delay	Pass criterion	Metrics recorded
M1	0%	0	0	0	100% of valid devices OPERATIONAL	Baseline time, frame counts
M2	10%	0.05	0.02	1 ms	100% OPERATIONAL	Retx ratio, time
M3	15%	0.10	0.05	2±1 ms	100% OPERATIONAL (**the demo configuration**)	Retx ratio, time, OOO and dup counts
M4	20%	0.10	0.05	5±2 ms	100% OPERATIONAL; median time ≤ 60 s (SM-1, SM-4)	All of the above
M5	30%	0.20	0.10	5±2 ms	≥ 95% OPERATIONAL (stress, P2)	Session failures/resumes
M6	50%	0.20	0.10	5 ms	No crash and no corrupted state; completion not required (P3)	Behaviour notes
M7	15% correlated 25% (netem)	10%	5%	5 ms	100% OPERATIONAL	Comparison with M3
M8	Asymmetric: 20% device→server only	0	0	0	100% OPERATIONAL	Retx per direction
M9	Asymmetric: 20% server→device only	0	0	0	100% OPERATIONAL	Retx per direction

For every run: **zero** duplicate DB rows, **zero** duplicate REGISTERs after a restart, and every artifact hash verified (invariants checked by `tests/check_invariants.sh`).

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7. Security test plan

Area	Cases	Technique
Authenticity	TC-AUTH-02/03/08/09/11/12/13, TC-DISC-07	A wrong model key (Invalid HMAC), an unknown model, no policy, a wrong fleet key, bit flips at every offset, a rogue server and a rogue OFFER
Replay	TC-AUTH-04..07, TC-AUTH-14, TC-DISC-08, TC-SESS-07, TC-SESS-09, TC-NAT-12	The built-in `--replay-test`; `peer_sim` replaying captured datagrams (`tcpdump` → `peer_sim --replay file.pcap`)
DB integrity	TC-AUTH-05 (Success not overwritten), TC-UPG-04	Hostile attempts against existing serials
Input handling	TC-UNIT-04, TC-SRV-07, TC-REL-08	Fuzzing and boundary datagrams (0, 43, 44, 1200, 1201, 65507 bytes)
DoS resistance	TC-AUTH-10, TC-SRV-08, TC-SRV-11, TC-NAT-12, TC-SESS-09	Floods, half-open exhaustion, the failed-record cap, peer-hijack attempts by replay, session teardown by HELLO
Crypto correctness	TC-UNIT-01/02/05/16	Known-answer tests, constant-time review
Secrets hygiene	Manual review	No session keys on disk (`grep` the state dir for the K_s hex from debug runs); keys zeroised on free

Known residual risks that are accepted and not tested as defects: no confidentiality, the shared model key, no forward secrecy (DESIGN §18.3). QA confirms that these are documented in the README (TC-DEL-06).

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8. Performance, scale and soak

Test	Load	Duration	Targets	Tooling
TC-PERF-01	50 devices (spawned by `run_scale.sh`), 15% loss	Until complete	All OPERATIONAL in ≤ 120 s; server RSS ≤ 64 MB; CPU < 1 core	`/usr/bin/time -l` (macOS) / `-v` (Linux)
TC-PERF-01b (P3, exploratory)	200 devices	Until complete	Record only	Same
TC-PERF-02	The 5-device demo, 20% loss, 20 runs	—	Median ≤ 60 s, p95 ≤ 90 s	Timestamps from the logs
TC-PERF-04	5 OPERATIONAL devices	1 h	No leaks, stable RSS	`leaks`, valgrind

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9. Log and deliverable verification

README checklist (TC-DEL-05..07). Each item must be present **and** must agree with the implemented behaviour:

#	README item	PRD ID	✓
1	Q1 How reliability is implemented over UDP	DEL-06	□
2	Q2 How replay attacks are prevented	DEL-06	□

#	README item	PRD ID	✓
3	Q3 Why IP fragmentation is avoided	DEL-06	□
4	Q4 How NAT affects the protocol design	DEL-06	□
5	Q5 What happens under packet loss	DEL-06	□
6	Overall architecture and control flow	DEL-07	□
7	UDP session management	DEL-07	□
8	Identity, authentication, registration	DEL-07	□
9	Config and firmware provisioning	DEL-07	□
10	Version upgrades and restarts	DEL-07	□
11	Known limitations and tradeoffs	DEL-07	□
12	Bonus: packet loss and out-of-order handling	DEL-08	□
13	Bonus: duplicate detection	DEL-08	□
14	Bonus: simulation method	DEL-08	□
15	Bonus Q1–Q4	DEL-08	□
16	Build and run instructions	DEL-05	□
17	Where the server and client logs are	DEL-09/10	□
18	Why "keepalives matter in real deployments" (PDF §4)	DEL-12	□
19	Walkthrough of the Appendix A example with real logs	FR-33	□
20	The DB dump, shown in the PDF example format	FR-31	□
21	Deployment prerequisite (outbound UDP 5555/5556 allowed) and NAT ownership; Q4 answer covers every row of DESIGN §21.1 Q4	FR-36	□
22	Configuration files and keys (server.json, device.json, policy.json, profiles), with an example	FR-34	□

Log checks (TC-LOG-01..04, TC-DEL-09) are automated in `tests/check_logs.sh`.

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10. Requirements traceability matrix

Status values: □ Not run · ✅ Pass · ❌ Fail · ⚠ Waived. Every row was □ at v0.1; the statuses below are from the M7 release run (README §17.1). ⚠ marks a requirement whose cases include ones that are open (§12.5) or waived; the other cases of that row pass.

10.1 Functional

PRD ID	DESIGN §	Test cases	Status
ZTP-FR-01	§6.1	TC-APX-01, TC-DISC-01/02, TC-E2E-01, TC-PDF-01	✅
ZTP-FR-02	§6.3	TC-APX-02, TC-DISC-03, TC-E2E-01, TC-PDF-01/04	✅
ZTP-FR-03	§4.3, §1.3	TC-APX-03, TC-E2E-01, TC-PDF-05, TC-SESS-08	✅
ZTP-FR-04	§7	TC-APX-03, TC-E2E-01, TC-SESS-01/04	✅
ZTP-FR-05	§5.2, §8.1	TC-APX-03, TC-E2E-01, TC-REL-01, TC-UNIT-03/06	✅
ZTP-FR-06	§8.3	TC-E2E-01, TC-REL-02/06/07, TC-SESS-02/03/05, TC-UNIT-07	✅
ZTP-FR-07	§8.4	TC-E2E-01, TC-REL-02/11, TC-UNIT-10	✅
ZTP-FR-08	§8.5	TC-E2E-01, TC-REL-05, TC-UNIT-09	✅
ZTP-FR-09	§8, §16	TC-APX-10, TC-E2E-01, TC-PERF-03, TC-REL-09	✅
ZTP-FR-10	§9.2, §9.5	TC-APX-04, TC-E2E-01, TC-NAT-01/02	✅
ZTP-FR-11	§9.3	TC-E2E-01, TC-NAT-03/04/05/07/08/12/13, TC-SESS-05	⚠ (TC-NAT-12 CHUNK_REQ half open; §12.5)
ZTP-FR-12	§10.2	TC-APX-05, TC-AUTH-01, TC-E2E-01	✅
ZTP-FR-13	§10.1–10.3	TC-APX-05, TC-AUTH-02/03/09/12, TC-E2E-01, TC-UNIT-02/16	⚠ (open: AUTH-03; §12.5)
ZTP-FR-14	§10.4	TC-AUTH-04/05/06/07/14/15, TC-E2E-01, TC-NAT-12, TC-SESS-09	⚠ (open: AUTH-04, AUTH-05, AUTH-06, AUTH-07, AUTH-14, AUTH-15, SESS-09; TC-NAT-12 CHUNK_REQ half open; §12.5)
ZTP-FR-15	§11.1	TC-APX-06, TC-E2E-01, TC-PDF-03, TC-PROV-01/02	✅
ZTP-FR-16	§11.1	TC-APX-07, TC-E2E-01, TC-PROV-03	✅

PRD ID	DESIGN §	Test cases	Status
ZTP-FR-17	§11.2	TC-APX-06, TC-E2E-01, TC-PROV-01/03/06	✅
ZTP-FR-18	§11.3	TC-APX-06, TC-E2E-01, TC-PROV-01/03/07/10, TC-UNIT-01	✅
ZTP-FR-19	§11.4	TC-E2E-01, TC-PROV-08/09/10/13, TC-UNIT-14/19, TC-UPG-10	✅
ZTP-FR-20	§5.5	TC-E2E-01, TC-PROV-04/05/12/13, TC-UNIT-19	✅
ZTP-FR-21	§12.1, §12.6	TC-APX-07, TC-E2E-01, TC-UNIT-12, TC-UPG-01/05/06	✅
ZTP-FR-22	§12.2	TC-APX-07/08, TC-E2E-01, TC-UPG-02	✅
ZTP-FR-23	§12.3	TC-APX-08, TC-AUTH-07, TC-E2E-01, TC-NAT-05/09/13, TC-PROV-09, TC-SESS-07, TC-SRV-10, TC-UPG-03/04/08/10/11	⚠ (open: AUTH-07, SESS-07; §12.5)
ZTP-FR-24	§13	TC-APX-09, TC-E2E-01, TC-UNIT-17, TC-VAL-01/02	✅
ZTP-FR-25	§13, §12.5	TC-APX-09, TC-E2E-01, TC-UPG-09, TC-VAL-01/03	✅
ZTP-FR-26	§13	TC-APX-09, TC-E2E-01, TC-VAL-01/04	✅
ZTP-FR-27	§13, §13.1	TC-APX-09, TC-E2E-01, TC-VAL-01/05	✅
ZTP-FR-28	§14.1	TC-E2E-02, TC-PERF-01, TC-SRV-01/09	✅
ZTP-FR-29	§14.3	TC-APX-05, TC-AUTH-09/12, TC-CFG-06, TC-E2E-02, TC-SRV-02, TC-UNIT-11, TC-UPG-06	✅
ZTP-FR-30	§14.4	TC-AUTH-03, TC-E2E-02, TC-PDF-02, TC-SRV-03, TC-UPG-04, TC-VAL-05	⚠ (open: AUTH-03; §12.5)
ZTP-FR-31	§14.5	TC-E2E-02, TC-PDF-02, TC-SRV-04	✅
ZTP-FR-32	§14.5	TC-E2E-02, TC-PDF-02, TC-SRV-05	✅

PRD ID	DESIGN §	Test cases	Status
ZTP-FR-33	§17, §17.4, §4.5	TC-APX-01/02/03/04/05/06/07/08/09/10, TC-E2E-01, TC-LOG-01, TC-PDF-01	✅
ZTP-FR-34	§25	TC-CFG-01/02/03/04/05/07/09/10, TC-UNIT-18/20	✅
ZTP-FR-35	§9.7, §25.3, §25.6	TC-CFG-06/08/10, TC-NAT-08, TC-UNIT-20	✅
ZTP-FR-36	§9.7, §9.8	TC-AUTH-13, TC-NAT-08/09/10/11	⚠ (open: AUTH-13; §12.5)

10.2 Bonus

PRD ID	DESIGN §	Test cases	Status
ZTP-BON-01	§6.2	TC-DISC-04/05/06/07/08	✅
ZTP-BON-02	§5.2, §8.1	TC-REL-01, TC-UNIT-03	✅
ZTP-BON-03	§8.3	TC-REL-02	✅
ZTP-BON-04	§8.4	TC-REL-03/10, TC-UNIT-08	✅
ZTP-BON-05	§8.6	TC-PROV-11, TC-REL-04/05, TC-SESS-02/03, TC-UNIT-09	✅
ZTP-BON-06	§16	TC-SIM-01/02/03/04, TC-UNIT-15	⚠ (waived: SIM-04)
ZTP-BON-07	§8.7	TC-REL-04	✅
ZTP-BON-08	§17.2	TC-LOG-02, TC-REL-03	✅

10.3 Non-functional

PRD ID	DESIGN §	Test cases	Status
ZTP-NFR-01	§4.2, §19	TC-DEL-01, TC-PDF-05, TC-UNIT-01	✅
ZTP-NFR-02	§5.5	TC-PROV-04/12	✅
ZTP-NFR-03	§8	TC-APX-10, TC-E2E-02, TC-NAT-07, TC-REL-09	✅
ZTP-NFR-04	§16.3	TC-PERF-02	✅
ZTP-NFR-05	§14	TC-AUTH-10, TC-PERF-01, TC-SRV-08/11	⚠ (open: AUTH-10, SRV-08, SRV-11; §12.5)
ZTP-NFR-06	§10, §18	TC-AUTH-08/11/13/14/15, TC-CFG-08/09, TC-DISC-07, TC-NAT-06/12, TC-SESS-06/09, TC-UNIT-05	⚠ (open: AUTH-08, AUTH-11, AUTH-13, AUTH-14, AUTH-15, SESS-09; TC-NAT-12 CHUNK_REQ half open; §12.5)

PRD ID	DESIGN §	Test cases	Status
ZTP-NFR-07	§5.6, §14.6	TC-CFG-04, TC-PERF-04, TC-REL-08, TC-SRV-07/08/11, TC-UNIT-04/18	⚠ (open: SRV-08, SRV-11; §12.5)
ZTP-NFR-08	§17	TC-LOG-03/04	✅
ZTP-NFR-09	§16.1	TC-UNIT-15	✅
ZTP-NFR-10	§15.3, §14.4	TC-SRV-06/10, TC-UNIT-13, TC-UPG-07/11	✅
ZTP-NFR-11	§19.1	TC-DEL-01/08	✅

10.4 Deliverables

PRD ID	DESIGN §	Test cases	Status
ZTP-DEL-01	§19	TC-DEL-02	✅
ZTP-DEL-02	§19.1	TC-DEL-01	✅
ZTP-DEL-03	§4.2	TC-DEL-03	✅
ZTP-DEL-04	§4.3	TC-PDF-05, TC-SESS-08	✅
ZTP-DEL-05	§19, §21.3	TC-DEL-04	✅
ZTP-DEL-06	§21.1	TC-DEL-05	✅
ZTP-DEL-07	§21.3	TC-DEL-06	✅
ZTP-DEL-08	§21.2, §21.3	TC-DEL-07	✅
ZTP-DEL-09	§17.3	TC-DEL-02/09	✅
ZTP-DEL-10	§17.3	TC-DEL-02/09	✅
ZTP-DEL-11	§14.5	TC-DEL-02, TC-PDF-02, TC-SRV-04	✅
ZTP-DEL-12	§9.6, §21.3	TC-DEL-10	✅

Coverage: 67 of 67 PRD requirements have at least one test case. No test case is orphaned: each one in §5 references at least one PRD ID.

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11. Entry and exit criteria, defect management, reporting

11.1 Entry criteria (per milestone)

- The milestone's code builds with `-Werror` on both platforms, and `make test` (L1) passes.
- The DESIGN sections concerned are frozen; the wire spec (§5) is frozen from M2 onward.
- The test hooks the milestone needs are available (DESIGN §19.2).

11.2 Exit criteria (release, M7)

1. **100% of P1 cases pass** on macOS and on Linux, in the release build and the ASan build.
2. ≥ 90% of P2 cases pass; every P2 failure has a signed waiver.
3. Impairment rows M1–M4 and M7–M9 pass for all seeds; M5 ≥ 95%.
4. Success metrics SM-1…SM-10 (PRD §3.3) met.
5. No open Sev-1 or Sev-2 defects.
6. The traceability matrix (§10) shows every requirement as ✅ or ⚠ (waived with sign-off).
7. The README checklist (§9) is complete.

11.3 Defect severity

Sev	Definition	Examples	Rule
1 Critical	Crash, data corruption, security bypass, or a required flow broken	Accepting a bad MAC; a duplicate REGISTER after a restart; ASan error	Blocks the milestone gate
2 Major	A requirement fails under some conditions	Fails at 20% loss for one seed; resume re-downloads everything	Must be fixed before release
3 Minor	A deviation that doesn't affect the outcome	Wrong log wording; misaligned dump columns	Fix or waive
4 Trivial	Cosmetic or documentation	Typo	Backlog

Every defect report includes: TC ID, platform, build variant, **seed and netsim flags**, the log excerpts, and the pcap if relevant.

11.4 Reporting

- Per milestone: a one-page test summary (passed, failed, blocked; new defects; risks).
- At release: `TEST_REPORT.md` with the traceability matrix filled in, the impairment matrix results (time and retx ratio per row), and the perf numbers.

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12. QA timeline, dependencies, assumptions, risks

The overall plan is owned by PRD §11. This is the QA view of it.

12.1 QA timeline (in parallel with development; start Mon 2026-11-16, the same calendar as PRD §11)

Window	QA activity	Effort
P0 (16–17 Nov)	Review PRD and DESIGN for testability; freeze this plan	1 d

Window	QA activity	Effort
M1 (18–20 Nov)	The config, policy, key and provisioned-config fixtures (§4.3); review the L1 tests; run the M1 gate set	1.5 d
M2 (23–26 Nov)	Build `peer_sim` (`--rogue-offer`, `--rogue-server`, `--fake-server`, `--bad-fleet-key`, `--flood`, `--replay`), the log checkers (`check_logs.sh` with the server matching rule, `check_invariants.sh`, `check_appendix_a.sh`) and `pcapcheck.sh`. They are built here because their first users are the M2 gate cases (TC-DISC-07, TC-SESS-06, TC-REL-08, TC-NAT-06, TC-CFG-08), and in M1 there was no session traffic to drive or check. Then run the M2 gate set (below), and seeded impairment rows M1–M4 on the session and reliability cases	4.5 d
M3 (27–30 Nov)	Run the M3 gate set: registration outcomes, replay and security cases, policy and DB	1.5 d
M4 (1–2 Dec)	Run the M4 gate set: transfers, many frames in flight, PROVISIONING-session cases	1 d
M5 (3–4 Dec)	Run the M5 gate set: everything that needs resume or the full flow (UPG with all crash points, VAL, kill -9 and server-restart resume, MTU fallback on CI, OPERATIONAL-state NAT cases, APX)	1.5 d
M6 (7 Dec)	Run the M6 gate set: SRV (including TC-SRV-10), PERF, E2E, PDF-02/05, LOG	1 d
M7 (8–9 Dec)	Full regression on both platforms, the whole matrix, soak, README checklist, test report	2 d
Total QA effort		**≈ 14 days**, overlapping development; about 2 days on the critical path after code freeze

Milestone gate sets. Each test runs at the earliest milestone that delivers everything it depends on. A milestone's exit criterion (PRD §11.1) is "its gate set passes, and every earlier gate set still passes". This table is the single source for those criteria.

Gate	Test cases (prerequisites in brackets)
M1	TC-UNIT-01..06, TC-UNIT-11..20, TC-CFG-01..05, TC-CFG-07 (codec, crypto, netsim, config and policy loaders, transfer bitmap and provisioned-config check; no reliability engine, which is built in M2)
M2	TC-UNIT-07..10 (reliability engine, simulated clock), TC-DISC-01..08, TC-SESS-01..04/06/08, TC-REL-01/02/03/08, TC-NAT-06/11, TC-SIM-01/02 (OOO half from M4), TC-CFG-08 (discovery, handshake, reliability, path probe; driven by HELLO, keepalive or `peer_sim` traffic)

Gate	Test cases (prerequisites in brackets)
M3	TC-AUTH-01..06/08/09/11..14, TC-REL-04, TC-REL-11 (needs a session that lives 60 s; M2 sessions are half-open and expire after `nat.half_open_ttl_s`), TC-CFG-09, TC-SRV-03/08/11 (registration, keys, policy, DB). **Open (2026-11-11):** TC-AUTH-03/04/05/06/08/11/13/14 and TC-SRV-08/11 wait for attack-simulation tooling (§12.5); the rest of the M3 gate passes
M4	TC-PROV-01..07/10/11, TC-REL-06/07, TC-SESS-09, TC-NAT-05/12 (transfer into the inactive slot, a PROVISIONING session, many frames in flight). **Open (2026-11-11):** TC-SESS-09 and the CHUNK_REQ-replay half of TC-NAT-12 wait for the replay tooling (§12.5); the rest of the M4 gate passes (`tests/m4_provision.sh`), and the §5.8 fallback (TC-PROV-13 (b), (c), locally) passes ahead of its M5 gate
M5	TC-UPG-01..11, TC-VAL-01..05, TC-SESS-05/07, TC-AUTH-07/10/15, TC-REL-05/09/10, TC-PROV-08/09/12/13, TC-NAT-01..04/07..10/13, TC-CFG-06/10, TC-APX-01..10, TC-PDF-01/03/04 (anything that completes the full flow or needs the resume handshake). **Status (2026-11-11): closed**, with the exceptions below. `tests/m5_upgrade.sh`: TC-UPG-01/02/03/05/07/08/09/10, TC-UPG-11 (a), TC-VAL-01/02/03/05, TC-CFG-10 (f), TC-NAT-10 (strict mode, Linux CI). `tests/m5_gate.sh`: TC-UPG-02 (fd count), TC-UPG-06, TC-UPG-11 (b), TC-VAL-04, TC-SESS-05, TC-REL-05/10, TC-PROV-09, TC-NAT-01..04/08/09, TC-PDF-01/03. `tests/check_appendix_a.sh`: TC-APX-01..10, TC-PDF-04 and TC-PROV-12 (CI `platform=linux`). `make longrun`: TC-NAT-07/13 (run 2026-11-11 on the Mac: TC-NAT-07, 86 keepalives while OPERATIONAL in 30 min, 0 session timeouts; TC-NAT-13, 280 keepalives in 10 min, 1 timeout (3.6×10^{-3} per cycle, consistent with DESIGN §9.3), resumed in 0.2 s, no re-registration). `tests/m4_provision.sh`: TC-PROV-08/13; unit tests: TC-CFG-06. **Exceptions:** TC-AUTH-07/10/15, TC-SESS-07 and the literal `nonce=984312` of TC-APX-05 wait for the open tooling (§12.5), as at M3 and M4; TC-UPG-04 and TC-REL-09 moved to the M6 gate (they are defined on the 5-device demo)
M6	TC-SRV-01/02/04..07/09/10, TC-PERF-01..04, TC-E2E-01/02, TC-PDF-02/05, TC-LOG-01..04, and from M5 TC-UPG-04 and TC-REL-09 (server hardening, dump, demo, scale). TC-UPG-04 checks the 5-device demo's dump, and TC-REL-09 runs that demo over impairment rows M2–M4: both need `scripts/run_demo.sh`, an M6 deliverable (moved 2026-11-11). **Status (2026-11-12): closed** with the exceptions below (CI green on b5b8bba, all jobs on both platforms). `scripts/run_demo.sh` (`make demo`, every push via `make test`): TC-SRV-01/04/05/06/09, TC-PDF-02, TC-UPG-04, TC-LOG-01..04 (`tests/check_logs.sh`), and the invariants (`tests/check_invariants.sh`). `tests/m6_server.sh`: TC-SRV-02/10. `tests/m6_fuzz.sh`: TC-SRV-07 (3 × 334,000 datagrams under ASan in CI, no report). `scripts/run_matrix.sh`: TC-REL-09 and TC-E2E-02 (rows M1–M4 × 10 seeds locally: every run and all valid devices OPERATIONAL; CI 3 seeds per push, 10 nightly) and TC-PERF-02 (row M4 × 20: median 3.9 s, p95 11.5 s against ≤ 60 / ≤ 90 s). `tests/check_appendix_a.sh`: TC-E2E-01. `scripts/run_scale.sh`: TC-PERF-01 (50 devices OPERATIONAL in 16–19 s, server peak RSS 5–8 MB, against 120 s / 64 MB; locally and in CI). `tests/m6_perf03.sh`: TC-PERF-03 (1.15 / 1.35 / 1.72 transmissions per frame at 10/20/30% loss against the model's 1.23 / 1.56 / 2.04). `scripts/run_soak.sh`: TC-PERF-04 PASS (1 h with 5 devices under caffeinate, no sleep in the window: every process's RSS fell from warm-up to end, e.g. server 1056 → 928 kB; `leaks` 0 on all 6 processes; 9155 keepalives, 0 session timeouts). `tests/check_pdf05.sh`: TC-PDF-05. CI row M7 (netem): 3/3 demos. **Exceptions:** the scenarios that need `--bad-key` and `--replay-test` (open tooling, §12.5; PRD R-15): the demo's ZTP-84BC19 row (TC-PDF-02 shows 4 of the PDF's 5 rows), the failure count in TC-SRV-05, the "Invalid HMAC" note, and ZTP-71DD08's failed_attempts = 1 (TC-E2E-02, QA §4.1). The demo detects the hooks and runs those scenarios when they exist

Gate	Test cases (prerequisites in brackets)
M7	TC-DEL-01..10, TC-SIM-03/04, the full impairment matrix (§6), and a full regression of every gate. **Scope added 2026-11-12 (owner):** (1) README covers PDF "README Requirements" Q1–Q5, Expected Submission §2 topics, the Bonus section (3 topics + B1–B4) and DEL-12; (2) the committed `logs/db_dump.txt` reproduces the PDF §9 dump, except the ZTP-84BC19 "Invalid HMAC" row, which needs `--bad-key` (PRD R-15) and is submitted under a waiver listed in the README; (3) committed logs: `logs/` (5-device demo: `server.log`, `device_<serial>.log`, `db_dump.txt`) and `logs/appendix_a/` (the single-device Appendix A journey, ZTP-93AF21, on 127.0.0.1:5555 on the Mac; CI runs it on 10.10.10.5); (4) a configuration reference for every file in `conf/` (keys, types, defaults, ranges); (5) a folder and file map with a one-line purpose and limitations; (6) every regression suite writes `results/<date-time>/<suite>/` (logs plus `summary.txt`, ignored by git), and one final full release run is copied to `evidence/` and committed; (7) CI evidence: a workflow step writes each job's summary as an uploaded report, and `evidence/ci/` records per-job results, run links and dates; (8) instructions for running the full regression and any single suite or case by hand, on the Mac and on Linux. `tests/run_all.sh` named in §3 is `make test`. **Status (2026-11-12): closed.** Release regression `make release` on b936db3 (README §17.1, `evidence/release-20261112/`): 23/23 steps PASS with no host sleep, the ASan build 16/16, matrix rows M1–M5, M8, M9 × 10 seeds 100% OPERATIONAL, TC-PERF-01..04, TC-NAT-07/13; CI green on 821d134 and a1c5bd3 (b936db3: tests green, one artifact upload failed on a storage error). TC-DEL-01 (clean build, no warnings) and TC-DEL-04 (README quick start on a fresh clone: `make`, `make demo` PASS) on b936db3. The §10 matrix is filled in: every requirement ✅ or ⚠. The release runs found and fixed: two harness bugs, three timing-sensitive demo checks, and a device defect (false PMTU black-hole step-down under loss, DESIGN §5.8). **Exceptions:** the 16 open-tooling cases (§12.5, PRD R-15) and TC-SIM-04 (waived)

12.2 QA dependencies

- QA-D1: the test hooks in DESIGN §19.2 (all gated by `--test-hooks`): `--first-session-id`, `--corrupt-chunk`, `--drop-seq`, `--dup-seq`, `--no-discovery`, `--bad-key`, `--replay-test`, `--reuse-nonce`, `--nonce`, `--crash-after-chunks`, `--crash-at`, `--restart-mode`, `--restart-delay`, `--rebind-after`, `--fail-validate`, `--hook-delete-config`, `--no-refetch`, `--hook-bad-fw-version`, plus the simulation flags `--sim-mtu` and `--nat-sim`. These must land with each milestone. **Status (2026-11-11):** `--bad-key`, `--replay-test`, `--reuse-nonce` and `--nonce` are not built yet (§12.5).
- QA-D2: **GitHub Actions** on https://github.com/heysunilvk/ztp-project (PRD D-09) as the Linux environment for netem (TC-SIM-03), broadcast (TC-DISC-05 (b)), valgrind and the gcc build. There's no local Linux host or VM, because of limited disk space on the Mac. CI is set up in M1.
- QA-D3: the Xcode CLT on the macOS host. Satisfied: verified on 2026-11-10 (PRD D-08).
- QA-D4: `tcpdump` permission on the test host (sudo), for the pcap-based cases.

12.3 QA assumptions

- Loopback timing is representative enough of LAN behaviour; WAN conditions are approximated with `--delay`.
- The test hooks don't change production code paths beyond the injection point (verified by code review).
- Seeds 1–10 give sufficient coverage of the impairment matrix; a failing seed is added to the regression set permanently.
- **NAT:** there is no NAT hardware. The in-code NAT emulator (DESIGN §9.8: idle-timeout expiry with a new source port, reboot flush, source-IP change, port blocking) is assumed to represent what matters to the protocol, which is that mappings expire and external addresses change. It does **not** model NAT *types* (symmetric vs endpoint-independent filtering), ALG payload rewriting, or real conntrack timing. These are covered by design argument (DESIGN T-09, PRD A-12) and, where possible, by the optional CI job with Linux network namespaces and `iptables MASQUERADE` (§14 item 2). Real customer NATs remain untested (§13).

- **NAT mapping lifecycle (DESIGN T-12, PRD A-14):** each assumed aspect has a test. Creation by the first outbound datagram and refresh by outbound traffic only: TC-NAT-01/08 (the emulator refreshes only on device TX). A changing external port: TC-NAT-05/09. A changing external IP: TC-NAT-10. A new mapping after a restart: TC-UPG-03 (the device re-binds its socket after `execv`, and the server logs the new peer address with no re-registration). Deletion without notice: TC-NAT-09. Not testable here: real NAT mapping types and real conntrack lifetimes (§13).
- **NAT timing in tests:** the emulator's timeouts are scaled to the `demo` profile (for example a 1.5 s idle timeout against a 2 s keepalive). The ratios, not the absolute values, are assumed to carry over to production (a 15 s timeout against a 20 s keepalive).
- **Deployment prerequisite (PRD A-09):** test environments allow outbound UDP on 5555/5556, except in TC-NAT-11, which blocks it on purpose.

12.4 Risks to testing

Risk	Mitigation
Timing-dependent flaky tests	Assert on counts and states; generous bounds; re-run with the same seed to separate a flake from a bug
A test hook masks real behaviour	Hooks are reviewed; at least one kill -9 is a *real* external kill (TC-PROV-08)
Platform-specific socket behaviour (broadcast, DF)	Run both platforms; known platform differences documented (DESIGN §6.2)
Limited test time at M7	Most cases are automated and run from M2 onward, so M7 is mainly regression
The Mac's idle sleep freezes an unattended local run (found in M6: an intermittent TC-REL-11 failure, "idle device sent 0 keepalives", matched `pmset` sleep entries to the second)	`make test`, `demo`, `scale`, `soak` and `longrun` hold a `caffeinate -i` assertion while they run; wrap a hand-started run the same way

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12.5 Open test tooling (recorded 2026-11-11, M3; extended at M5)

The M3 code these tests exercise is built and passes its other tests, and the logic behind each one is in place. What is missing is the tooling that **generates the hostile or malformed traffic** to test it end to end: the device test hooks and `peer_sim` modes below. They weren't built with M3, and the owner takes them on. Until they exist, the listed cases are recorded as open exceptions to the M3 gate (§11.1), not as passes.

Missing tooling	Unblocks	Logic already built (and how it is covered today)
Device hook `--bad-key` (corrupt K_mfg; K_fleet stays correct)	TC-AUTH-03; demo device ZTP-84BC19 (PRD §5.2 S2)	BAD_MAC NAK echoing NONCE_REG; DB "Invalid HMAC"; device retries up to `auth.register_attempts`, then FAILED (code review; DB rules unit-tested in `tests/test_db.c`)
Device hook `--replay-test`	TC-AUTH-04, TC-AUTH-05; demo device ZTP-71DD08 (S3)	Duplicate drop in the reliability window (TC-NAT-12, TC-REL-04 pass); BAD_MAC for a frame under another session's K_s; a failure never overwrites a Success row (unit-tested)
Device hooks `--reuse-nonce`, `--nonce N`	TC-AUTH-06; Appendix A nonce 984312 (TC-APX-05)	REGISTER nonce cache → REPLAY NAK (code review)
A tool that bit-flips each byte of a captured REGISTER (and CHUNK_DATA from M4)	TC-AUTH-08	MAC verification on every frame (TC-UNIT-03 flips every bit of a frame; TC-NAT-06 passes)

Missing tooling	Unblocks	Logic already built (and how it is covered today)
`peer_sim` rogue server (bad-MAC HELLO_ACK)	TC-AUTH-11	Device verifies HELLO_ACK under K_boot with the Nd echo (code review)
`peer_sim` wrong fleet key	TC-AUTH-13	Silent drop and `mac_fail` on the server; blocked-path hint on the device (TC-NAT-11 passes)
`peer_sim` forged and replayed REGISTER_NAK	TC-AUTH-14	Device accepts a NAK only under K_boot with the outstanding NONCE_REG (code review)
`peer_sim` half-open load (300 HELLOs)	TC-SRV-08	`limits.max_half_open` → `SERVER_BUSY`; half-open TTL (TTL expiry passes in the M2 tests)
`peer_sim` fresh HELLOs for a serial with a live session, and a replay of the device's original HELLO	TC-SESS-09 (M4)	A HELLO never supersedes a registered session (§7.3, §14.2: only a verified REGISTER or resume does); cached HELLO_ACK for a repeated (serial, Nd) (TC-SESS-02/03 pass)
`peer_sim` replay of a captured CHUNK_REQ from another port	TC-NAT-12, CHUNK_REQ half (M4; the KEEPALIVE half passes since M2)	The same duplicate and peer-update rule as the KEEPALIVE case
`peer_sim` failing-registration load (20,000)	TC-SRV-11	Failed-row cap, oldest evicted, Success rows never evicted (unit-tested at 1,000 rows over a cap of 100)
`peer_sim` replay of a captured resume HELLO (M5)	TC-AUTH-07	Resume counter must be newer than the last accepted one → `RESUME_REJECTED`; the live session is untouched. The device's retry after one `RESUME_REJECTED`, and its fresh registration after a second, are code review only: no test makes the server reject a resume yet
`peer_sim` replay of an epoch-0 frame after a resume (M5)	TC-SESS-07	Both sides drop a frame from another epoch before the MAC check and count it as `rx_stale` (code review)
`peer_sim --flood` (HELLO flood from one IP) (M5)	TC-AUTH-10	Per-IP token bucket (§10.5) and the half-open cap (code review)
`peer_sim` replay of captured terminal ERRORs to another device (M5)	TC-AUTH-15	The device accepts a terminal ERROR only if it echoes the Nd/Nd′ of its pending HELLO or resume, and RESUME_REQUIRED only for a seq it sent in the current epoch, once per seq (code review)

Other QA tooling not yet built under its planned name (no gate case is blocked by it):

- The log checkers: `tests/check_appendix_a.sh` (built at M5), `tests/check_logs.sh` and `tests/check_invariants.sh` (built at M6, run by `scripts/run_demo.sh` on every demo and matrix run). The milestone test scripts keep checking their own lines inline.
- The CI jobs `impair-matrix`, `impair-netem`, `valgrind`, `mtu` and `nat-real` (DESIGN §19.4 status): M6, with the demo.

When a tool lands, its cases move back into the gate and run in CI like the others. Anything still open at M7 is listed in the README's "Test coverage and known gaps" section (DESIGN §21.3).

13. Caveats: what is not or cannot be tested

Gap	Why	Mitigation / substitute
Real NAT devices (port-restricted, symmetric, CGNAT)	No NAT hardware in the lab	Rebinding simulation (TC-NAT-05), the NAT emulator (TC-NAT-08..10), idle-timeout logic (TC-NAT-03/04), and the optional CI `nat-real` job (a Linux conntrack MASQUERADE NAT). Optional: a Linux `iptables MASQUERADE` namespace test (§14)

Gap	Why	Mitigation / substitute
Real power loss and flash behaviour	No hardware	Crash injection at every commit point (TC-UPG-07); atomic-write unit tests
Real wireless or satellite loss patterns	No RF test bed	Correlated loss with netem on CI (M7); a range of seeds
Interactive Linux debugging	No local Linux host	Reproduce with the same seed on the Mac (netsim is deterministic); CI uploads logs and pcaps as artifacts; a temporary `tmate` debug step in CI can be enabled if needed
Long-term fleet scale (10k+ devices)	Out of scope	50-device test plus a 200-device exploratory run; design analysis (DESIGN §14.1)
Cryptographic strength	Not a QA activity	Standard primitives with known-answer tests; design review
Confidentiality	Not implemented by design	Documented limitation (DESIGN §18.3)

14. QA enhancement suggestions

1. **Coverage-guided fuzzing** (libFuzzer or AFL++) of `ztp_decode()` and the TLV parser, run in CI for every change.
2. **Network-namespace NAT lab** on Linux (`ip netns` + `iptables MASQUERADE` with short UDP conntrack timeouts) for realistic NAT tests.
3. **Property-based tests** of the reliability engine: arbitrary interleavings of loss, reorder, duplication and delay, with the invariants "exactly once, in order, eventually complete when loss < 100%".
4. **Extend the CI matrix.** The base matrix (Ubuntu and macOS × release and ASan × rows M1–M4) is already in scope (DESIGN §19.4). Next steps: nightly runs of all 10 seeds on every row, and badges in the README.
5. **Code coverage** (`gcov` / `llvm-cov`) with a target of ≥ 85% lines in `common/` and ≥ 75% overall.
6. **A Wireshark Lua dissector** for the ZTP header, which makes pcap review and defect reports much faster.
7. **Chaos mode in the demo**: random kill -9 of devices and the server during runs, checked against the invariants.

15. PDF conformance matrix

This is the clause-by-clause check that the test plan covers **every point in the assignment PDF**. Each row gives the PRD requirement (PRD Appendix: PDF coverage map), the DESIGN section, and the test cases that must pass.

#	PDF section and clause	PRD ID	DESIGN §	Test cases
1	Overview: ZTP over UDP transport	FR-03	§1.3, §5	TC-SESS-08, TC-PDF-05
2	Overview: not a REST or API exercise	FR-03, NFR-01	§1.3	TC-PDF-05
3	Scope: language C or Python	NFR-01	§1.3	TC-DEL-01, TC-PDF-05

#	PDF section and clause	PRD ID	DESIGN §	Test cases
4	Scope: working code and a README	DEL-01..08, DEL-12	§19, §21	TC-DEL-02, TC-DEL-04..07, TC-DEL-10
5	Scope: correctness and fundamentals over UI or packaging	— (scope)	§1.3	TC-PDF-05 (no UI or HTTP code)
6	Problem statement: unconfigured device to OPERATIONAL	FR-01..27	§4.5	TC-E2E-01, TC-APX-01..10
7	§1 Generate or load a soft serial	FR-01	§6.1	TC-DISC-01/02, TC-APX-01
8	§1 Static IP discovery	FR-02	§6.3	TC-DISC-03, TC-APX-02, TC-PDF-04
9	§1 Broadcast or multicast discovery (bonus)	BON-01	§6.2	TC-DISC-04..08
10	§1 Example output (`Soft serial generated`, `Provisioning server found`)	FR-33	§6.1, §6.3, §17.4	TC-PDF-01
11	§2 All communication over UDP	FR-03	§5.1	TC-SESS-08
12	§2 Session handshake	FR-04	§7	TC-SESS-01/02/03, TC-APX-03
13	§2 Sequence numbers	FR-05	§5.2, §8.1	TC-UNIT-03/06, TC-REL-01
14	§2 Timeouts	FR-06	§8.3	TC-UNIT-07, TC-REL-02, TC-SESS-05
15	§2 Retransmission logic	FR-06	§8.3, §8.10	TC-REL-02/07, TC-SESS-02
16	§2 Session identifier	FR-04	§7.2	TC-SESS-01/04, TC-APX-03
17	§3 Acknowledgements	FR-07	§8.4	TC-UNIT-10, TC-REL-02/11
18	§3 Duplicate detection	FR-08	§8.5	TC-UNIT-09, TC-REL-05
19	§3 Retransmissions on timeout	FR-06	§8.3	TC-REL-02
20	§3 Simulated packet loss of 10–20%	FR-09, NFR-03	§8, §16	TC-REL-09, matrix rows M2–M4, TC-APX-10, TC-E2E-02
21	§4 Device may be behind NAT or a firewall (owned by the customer or ISP, may change; NAT parameters configurable)	FR-11, FR-34..36	§4.6, §9.1, §9.4, §9.7, §9.8, §25	TC-NAT-05/06/08/09/10/11/12/13, TC-CFG-06/08
22	§4 Periodic keepalive messages	FR-10	§9.2	TC-NAT-01/02, TC-APX-04
23	§4 Session timeout detection	FR-11	§9.3, §9.5	TC-NAT-03/04/07
24	§4 README: why this matters for real deployments	DEL-12	§9.6	TC-DEL-10
25	§5 REGISTER carries soft serial, metadata, nonce	FR-12	§10.2	TC-AUTH-01, TC-APX-05
26	§5 HMAC verification	FR-13	§10.1, §10.3	TC-UNIT-02/16, TC-AUTH-02/03/08/09/12/13
27	§5 Replay protection	FR-14	§10.4	TC-AUTH-04/05/06/07/14/15, TC-NAT-12, TC-SESS-09

#	PDF section and clause	PRD ID	DESIGN §	Test cases
28	§6 Configuration payload (JSON or YAML)	FR-15	§11.1	TC-PROV-01/02, TC-PDF-03
29	§6 Firmware blob (mock)	FR-16	§11.1	TC-PROV-03
30	§6 Transfer in chunks	FR-17	§11.2	TC-PROV-01/03/06, TC-APX-06
31	§6 Integrity hash	FR-18	§11.3	TC-PROV-07/10
32	§6 Resume transfers after failure	FR-19	§11.4, §11.5	TC-PROV-08/09/13, TC-UPG-10, TC-UNIT-19
33	§6 No reliance on IP fragmentation	FR-20	§5.5, §5.8	TC-PROV-04/05/12/13
34	§7 Outdated firmware: simulate upgrade	FR-21	§12.1, §12.6	TC-UPG-01/05/06, TC-APX-07
35	§7 Restart the device process	FR-22	§12.2	TC-UPG-02, TC-APX-07/08
36	§7 Resume without duplicate registration	FR-23	§12.3, §14.4	TC-UPG-03/04/08/10/11, TC-SRV-10, TC-PROV-09, TC-APX-08
37	§8 Validate configuration availability	FR-24	§13, §13.1	TC-VAL-01/02, TC-APX-09
38	§8 Validate firmware version	FR-25	§13	TC-VAL-03, TC-UPG-09, TC-APX-09
39	§8 Validate session health	FR-26	§13	TC-VAL-04
40	§8 Transition to OPERATIONAL	FR-27	§13.1, §15.1	TC-VAL-01/05, TC-APX-09
41	§9 Multiple devices concurrently	FR-28	§14.1, §14.8	TC-SRV-01/09, TC-PERF-01
42	§9 Accept and process registrations at the same time	FR-28	§14.8, §14.9	TC-SRV-09, TC-AUTH-10
43	§9 Determine and assign firmware and config per device	FR-29	§14.3	TC-SRV-02, TC-UPG-06, TC-AUTH-09
44	§9 Track: device ID, timestamp, firmware, config version, outcome	FR-30	§14.4	TC-SRV-03
45	§9 In-memory or file-backed store	FR-30, NFR-10	§14.4	TC-SRV-06
46	§9 Dump the database	FR-31	§14.5	TC-SRV-04
47	§9 Summary: successes, failures, versions per device	FR-32	§14.5	TC-SRV-05
48	§9 "Example of a Dump from Server DB" (format and rows)	FR-31	§14.5	TC-PDF-02
49	README Q1 reliability over UDP	DEL-06	§21.1	TC-DEL-05
50	README Q2 replay prevention	DEL-06	§21.1	TC-DEL-05

#	PDF section and clause	PRD ID	DESIGN §	Test cases
51	README Q3 why IP fragmentation is avoided	DEL-06	§21.1	TC-DEL-05
52	README Q4 how NAT affects the design	DEL-06	§21.1 (with §9.7)	TC-DEL-05
53	README Q5 behaviour under packet loss	DEL-06	§21.1	TC-DEL-05, TC-PERF-03
54	Notes: reasoning and tradeoffs	DEL-07	§3, §22	TC-DEL-06
55	Appendix A scenario: no configuration, certificates or credentials	(A-02)	§3.1	TC-DISC-01 (empty state), TC-PDF-01
56	Appendix A device information (ZTP-93AF21, EDGE-GW-100, v1.0)	FR-01	App. A	TC-APX-01
57	Appendix A Step 1 Device boot	FR-01, FR-33	§6.1	TC-APX-01
58	Appendix A Step 2 Server discovery (10.10.10.5:5555)	FR-02	§6.3	TC-APX-02, TC-PDF-04
59	Appendix A Step 3 UDP session (HELLO, HELLO_ACK, session_id)	FR-04, FR-05	§7.1	TC-APX-03
60	Appendix A Step 4 NAT keepalive	FR-10	§9.2	TC-APX-04
61	Appendix A Step 5 Secure registration (fields, nonce, tenant)	FR-12, FR-13	§10.2	TC-APX-05
62	Appendix A Step 6 Configuration provisioning (example config, 3 chunks)	FR-15, FR-17, FR-18	§11	TC-APX-06, TC-PDF-03
63	Appendix A Step 7 Firmware upgrade (v1.0 → v1.1)	FR-21, FR-22	§12.1	TC-APX-07
64	Appendix A Step 8 Resume after restart	FR-23	§12.3	TC-APX-08
65	Appendix A Step 9 Validation and OPERATIONAL	FR-24..27	§13	TC-APX-09
66	Bonus assumptions: 10–20% loss, reordering, duplicates	NFR-03, BON-04/05	§8	TC-REL-05/09/10, TC-APX-10
67	Bonus req. 1: sequence numbers on every message	BON-02	§5.3	TC-REL-01
68	Bonus req. 2: ACKs and retransmission	BON-03	§8.3	TC-REL-02
69	Bonus req. 3: idempotent processing	BON-05	§8.6	TC-REL-04/05, TC-PROV-11, TC-SESS-02
70	Bonus req. 4: packet loss simulation in code	BON-06	§16	TC-SIM-01..04, TC-UNIT-15
71	Bonus sample packet flow diagram	BON-04	§8.8.1	TC-REL-03
72	Bonus out-of-order example (expected 2, received 3, buffer)	BON-04	§8.4	TC-REL-03, TC-UNIT-08
73	Bonus sample log output (4 lines)	BON-08	§17.2	TC-LOG-02, TC-REL-03/04

#	PDF section and clause	PRD ID	DESIGN §	Test cases
74	Bonus README Q1–Q4	DEL-08, BON-07	§21.2	TC-DEL-07
75	Expected Submission 1: implements ZTP	DEL-01	§19	TC-DEL-02, TC-E2E-02
76	Expected Submission 1: runnable or buildable as submitted	DEL-02	§19.1	TC-DEL-01
77	Expected Submission 1: device and server logic separated	DEL-03	§4.2	TC-DEL-03
78	Expected Submission 1: UDP sockets for all provisioning traffic	DEL-04	§4.3	TC-SESS-08
79	Expected Submission 1: build and run instructions	DEL-05	§19	TC-DEL-04
80	Expected Submission 2: README minimum topics (6)	DEL-07	§21.3	TC-DEL-06
81	Expected Submission 2: README bonus topics (3)	DEL-08	§21.3	TC-DEL-07
82	Expected Submission 3: server logs	DEL-09	§17.3	TC-DEL-02/09
83	Expected Submission 3: client logs	DEL-10	§17.3	TC-DEL-02/09

Result: all 83 PDF clauses map to a design section and at least one test case. 81 of them map to a PRD requirement ID. The other two map to a documented scope or assumption decision instead: row 5 ("correctness over UI or packaging") to PRD §10.1, and row 55 (Appendix A "no stored credentials") to assumption A-02 and DESIGN §3.1.
